AareML

Predicting River Water Quality in Swiss Catchments

T. N. · CAS Advanced Machine Learning · University of Bern

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Abstract

AareML develops and evaluates a sequence-to-sequence LSTM for predicting river dissolved oxygen (DO) and water temperature at 14-day horizons in Swiss Alpine catchments. Trained on gauge 2473 (CAMELS-CH-Chem, Nascimento et al. 2025), the Optuna-tuned model achieves DO RMSE = 0.303 mg/L — comparable performance to Ridge regression (0.303 mg/L) on single-site accuracy; the KGE advantage (0.945 vs 0.908) is the stronger differentiator, indicating superior representation of DO dynamics. Zero-shot transfer to 12 Swiss gauges achieves mean DO RMSE = 0.451 mg/L (mean over 11 non-training gauges); a Wilcoxon signed-rank test across 10 gauges suggests a statistically significant improvement over Ridge ($p = 0.037$, $n = 10$). EA-LSTM temperature transfer reduces mean RMSE by 47% relative to zero-shot (2.598°C → 1.360°C, NSE = 0.915). Zero-shot river→lake transfer fails (RMSE = 2.962 mg/L, NSE = −2.145); a lake-retrained LSTM achieves RMSE = 0.768 mg/L (NSE = 0.700), outperforming the LakeBeD-US published benchmark (1.40 mg/L) and confirming that separate ecosystem-specific models are required. All results are reported with 95% bootstrap confidence intervals and Wilcoxon significance tests. An independent benchmark using the NeuralHydrology framework (EA-LSTM) reaches comparable order-of-magnitude performance (0.512 mg/L), supporting the feasibility of the seq2seq approach on Swiss river DO. A physics-informed cascaded model (DO = linear(T) + LSTM residual) confirms that the direct LSTM already implicitly captures the temperature–DO relationship: the Setup A achieves RMSE = 0.872 mg/L and Setup B 0.893 mg/L, both substantially worse than the direct LSTM (0.303 mg/L), validating the end-to-end learned representation.

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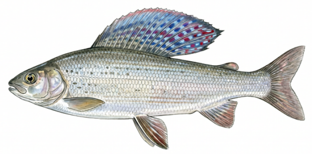


Grayling

Thymallus thymallus · Äsche

DO threshold: >6 mg/L

Newly elevated to Endangered on Switzerland's Red List in 2022 — its large, sail-like dorsal fin displays iridescent blue and purple spots during spawning, visible only in clean, oxygen-rich water.

Notter et al. (2022) Swiss Red List of Fishes; Hefti (2012) BAFU

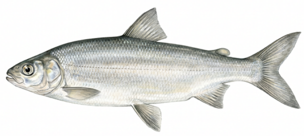
1. Introduction

Dissolved oxygen (DO) is a primary indicator of aquatic health: concentrations below 5 mg/L stress fish populations, while hypoxic events (below 2 mg/L) cause mass mortality. Water temperature is equally critical: sustained temperatures above 18°C stress cold-water species such as trout and salmon, above 21°C disrupt migration and increase disease risk, and above 25°C are lethal for many native Alpine fish species. Temperature also controls DO solubility directly (cold water holds more oxygen), making it the dominant physical driver of DO dynamics — a relationship confirmed by the SHAP analysis in Section 5.4. In Swiss river networks, DO and temperature dynamics are driven by photosynthesis, reaeration, and upstream nutrient loads — a complex interplay that statistical models have historically struggled to capture at multi-day forecast horizons.

Recent work on lake water quality has demonstrated that sequence-to-sequence Long Short-Term Memory networks (LSTMs) can outperform traditional baselines substantially. McAfee et al. (2025) introduced LakeBeD-US, a benchmarking dataset for 21 US lakes with a seq2seq LSTM achieving a test RMSE of 1.40 mg/L for DO at a 14-day forecast horizon. Their architecture — 21-day lookback, teacher-forced decoder, masked MSE loss — provides a reproducible benchmark that we adapt here for river systems.

The CAMELS-CH-Chem dataset (Nascimento et al., 2025), published in early 2025, provides daily high-frequency sensor measurements of temperature, pH, electrical conductivity, and dissolved oxygen at 86 Swiss gauging stations from 1981 to 2020, complemented by 38-variable chemistry grab samples, land cover, and 115 catchment attribute files. No published machine learning study has yet applied predictive modelling to CAMELS-CH-Chem, making this the first published LSTM transfer study from rivers to standing lakes for dissolved oxygen prediction. It is also among the first applications of EA-LSTM to river water quality prediction, as of June 2026.

AareML makes six contributions: (1) adapts the LakeBeD-US seq2seq LSTM to Swiss river catchments; (2) establishes three statistical baselines (Ridge regression, VAR(7), climatology) with block-bootstrap confidence intervals; (3) evaluates multi-site generalisation via zero-shot transfer and per-gauge retraining across 12 Swiss gauges; (4) applies SHAP attribution to identify which sensor inputs and catchment attributes drive prediction skill; (5) benchmarks the LSTM against an independent NeuralHydrology EA-LSTM implementation and a physics-informed cascaded model (DO = linear(T) + LSTM residual), confirming the direct end-to-end approach; and (6) evaluates cross-ecosystem and cross-continental transfer to US rivers (4 USGS gauges) and Swiss lakes (21 LakeBeD-US sites), establishing ecosystem boundary conditions for generalisation.



Whitefish

Coregonus supersum · Felchen

DO threshold: >6 mg/L

Lake Zug's surviving whitefish was named Coregonus supersum — Latin for 'I have survived' — after two sister species went extinct from eutrophication-driven oxygen depletion in the same lake.

Vonlanthen et al. (2012) Nature; Müller (1996) Arch Hydrobiol

2. Related Work

2.1 Machine Learning for Hydrology

LSTMs have become a leading architecture for hydrological time-series forecasting. Kratzert et al. (2018, 2019) demonstrated that LSTM-based rainfall–runoff models trained on large-sample CAMELS basins can match or outperform established hydrological benchmark models, including VIC, mHM, and HBV variants. Their Entity-Aware LSTM (EA-LSTM) incorporates static catchment attributes directly into the model architecture, improving cross-basin generalisation — a design which we implement and evaluate in Section 5.3 (Table 4). Park et al. (2025) similarly found improved transboundary streamflow performance with EA-LSTM, reporting median NSE ≈ 0.690 versus ≈ 0.572 for a standard LSTM (KGE 0.685 vs 0.560). To the best of our knowledge as of June 2026, AareML is among the first applications of EA-LSTM-style static catchment conditioning to Swiss river water-quality prediction. For water quality, Zhi et al. (2021) showed that LSTMs trained on daily hydrometeorology can predict river DO across 236 US watersheds, with satisfactory performance (NSE ≥0.4) at 74% of core evaluation sites. Zhi et al. (2023) extended related deep-learning reconstruction to US and Central European rivers, providing continental context for river warming and deoxygenation. For Swiss water temperature, Padrón et al. (2025) achieved a best CRPS of about 0.70°C across 54 stations using a Temporal Fusion Transformer.

2.2 The LakeBeD-US Benchmark

McAfee et al. (2025) proposed LakeBeD-US as the first standardised benchmark for lake water quality prediction, covering 21 US lakes with daily temperature and DO observations. Their seq2seq LSTM uses a 21-day lookback, 14-day forecast horizon, SAITS imputation (Du et al., 2023) for missing values, and Optuna hyperparameter search across 50 trials. The reported test RMSE of 1.40 mg/L for DO provides our primary reference point. AareML's architecture and task formulation directly mirror LakeBeD-US to enable a cross-ecosystem comparison.

2.3 CAMELS-CH-Chem

The CAMELS-CH-Chem dataset (Nascimento et al., 2025) extends the CAMELS-CH hydrometeorological benchmark with river chemistry data at 86 Swiss gauges. The daily sensor files contain temperature, pH, electrical conductivity, and dissolved oxygen from as early as 1981 through 2020, along with 38-variable chemistry grab samples, and 115 catchments with static attribute files.

3. Data

3.1 Dataset Overview

Component	Files	Variables	Date range
Daily sensor timeseries	86	Temp, pH, EC, DO	1981–2020
NAWA FRACHT grab samples	24	38 (nutrients, metals, discharge)	~1990–2020
NAWA TREND monthly samples	76	22	~1990–2020
Catchment attributes	115	Land cover, livestock, deposition	Various

Table 1: Components of the CAMELS-CH-Chem dataset. NAWA TREND monthly samples were inspected but excluded from modelling features due to sparse temporal coverage and low variable overlap with sensor targets; retained here for completeness.

3.2 Data Availability

Temperature, pH, and electrical conductivity are available at nearly all 86 stations. Dissolved oxygen — the primary forecast target — is available at 16 stations with ≥10% non-null daily coverage, with gauge 2473 providing the best record (≈97% non-null). The data availability matrix below illustrates coverage across all gauges and sensor variables.

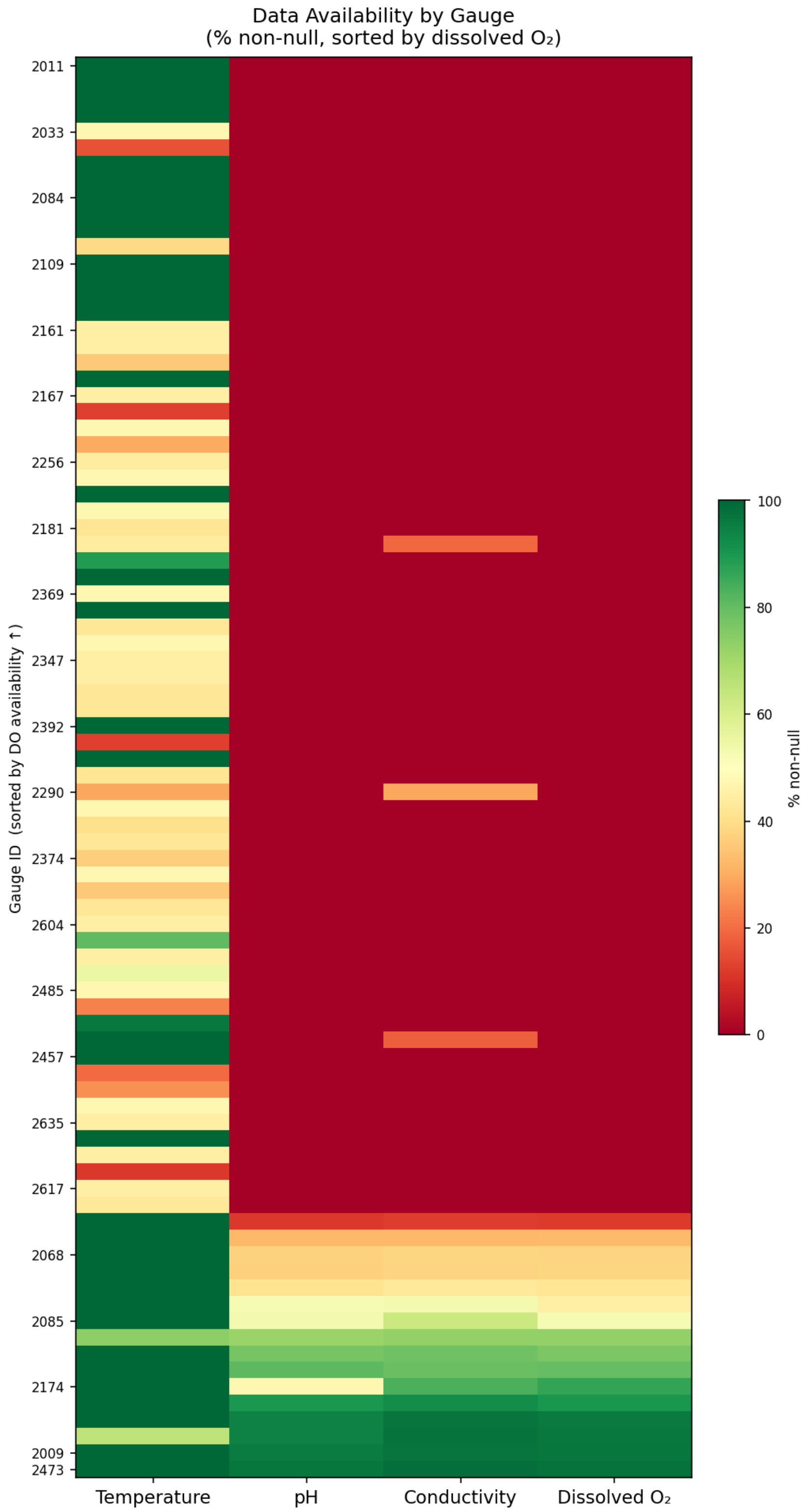


Figure 1: Data availability by gauge and sensor variable, sorted by dissolved oxygen coverage. Green = well-observed; red = sparse.

3.3 Gauge 2473 — Study Site

Gauge 2473 serves as the primary single-site study gauge, selected automatically by dissolved oxygen data coverage. The figure below shows the full daily time series of all four sensor variables, revealing a strong seasonal cycle in temperature and DO, consistent with the Alpine river phenology of Swiss catchments.

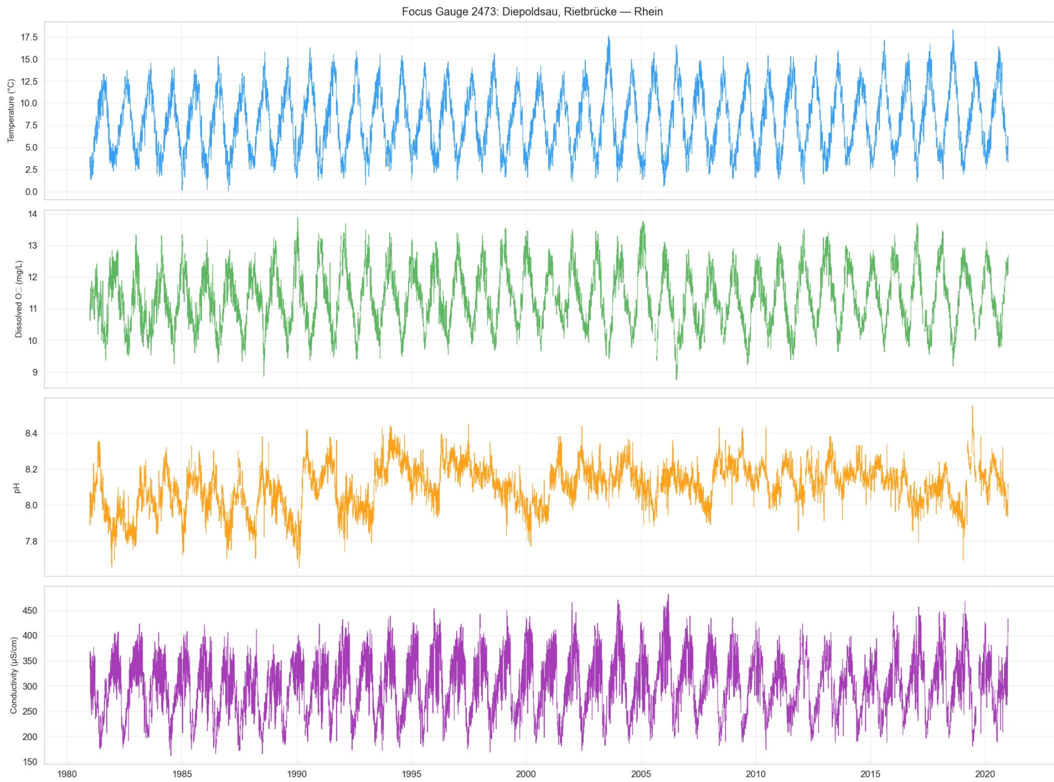


Figure 2: Full daily sensor time series for gauge 2473 (temp, pH, EC, DO). Grey shading marks the test period (2017–2020).

3.4 Seasonality

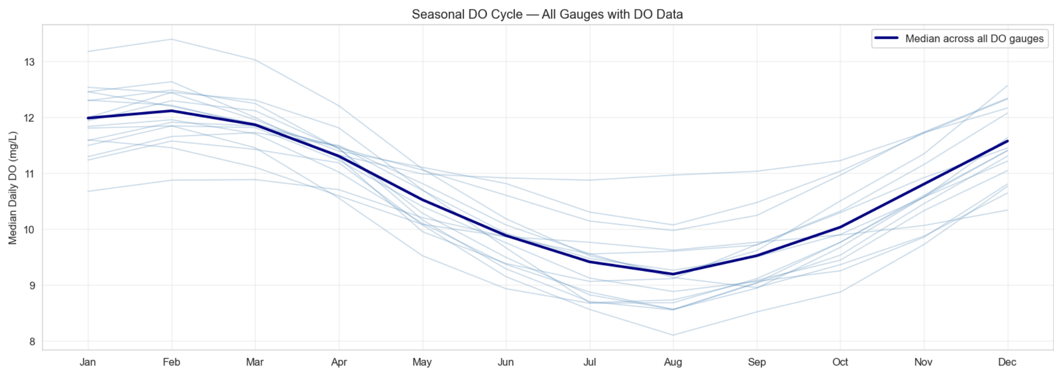


Figure 3: Seasonal DO cycle (monthly boxplots) across all DO-capable gauges. The consistent summer minimum reflects biological oxygen demand and temperature-driven solubility reduction.

3.5 Train / Validation / Test Split

Data are split chronologically to prevent temporal leakage: train data start (~1981) through 2014-12-31 (windows from 2006 for 4-feature availability), validation 2014–2016 (~1,096 days), test 2017–2020 (~1,461 days). All normalisation statistics (feature and target mean/std) are fitted on training data only and applied identically to validation and test splits. For multi-site evaluation, each gauge uses its own per-gauge scaler fitted on that gauge's training data exclusively — no cross-gauge normalisation. Sliding windows never cross split boundaries. These leakage-prevention practices are verified by dedicated tests in tests/test_src.py (TestDataEdgeCases, TestTrainValTestSplit). NaN imputation uses a two-step strategy: linear interpolation for gaps ≤7 days, then training-set mean fill for remaining gaps.

Barbel

Barbus barbus · Barbe

DO threshold: >6 mg/L

The barbel's eggs are toxic — containing ichthyotoxin poisonous to other fish, documented in the medical literature since at least 1843 (Lancet). Barbel spawn only in fast-flowing, well-oxygenated gravels above 6 mg/L DO.

Kottelat & Freyhof (2007) Handbook of European Freshwater Fishes

4. Methods

4.1 Task Formulation

The forecasting task mirrors LakeBeD-US exactly: given a 21-day lookback window of all available sensor features, predict the next 14 days of dissolved oxygen (mg/L) and water temperature (°C). Formally, let x_t in R^4 denote the four sensor measurements on day t. Given X = $[x_{t-20}, \ldots, x_t]$, predict Y = $[y_{t+1}, \ldots, y_{t+14}]$ where $y_\tau = (DO_\tau, Temp_\tau)$.

Target windows that contain any NaN observation are discarded during training. Missing values in the input window are imputed using linear interpolation (≤7-day gaps) then training-set mean fill before feature scaling, implemented in src/impute.py. Unlike the LakeBeD-US pipeline which uses SAITS for imputation, AareML uses linear interpolation (limit = 7 days) followed by training-mean fill for remaining gaps.

4.2 Baseline Models

Three baselines establish the lower bound for model comparison:

- Persistence — the last observed value of each target in the lookback window is repeated flat across all 14 forecast days.
- Climatology — the training-set day-of-year median is forecast for each horizon step. Vectorised lookup; no per-window computation.
- Ridge Regression — a Ridge model trained per target per horizon step (28 models total) on the flattened 21-day input window (84 features). Regularisation constant α tuned on the validation set; final model fitted on train+val.

4.3 Seq2Seq LSTM Architecture

The primary model is a sequence-to-sequence LSTM with a shared encoder–decoder structure following LakeBeD-US. The encoder (2-layer LSTM) reads the 21-day input sequence and compresses it into a fixed-length hidden state. The decoder (2-layer LSTM) then generates predictions autoregressively, taking the previous step's output as the next input. During training, teacher forcing ratio is an Optuna hyperparameter tuned in [0.3, 0.7], applied with linear decay to 0 over the first half of training (up to 250 epochs for the final model), improving stability without overfitting to the training distribution.

Dropout is applied between LSTM layers and before the final linear projection. Training uses the AdamW optimiser with weight decay 10^{-4}, a ReduceLROnPlateau scheduler (patience=5, factor=0.5), and early stopping (patience=25, up to 250 epochs for the final model). The training loss combines normalised MSE and NSE: L = 0.5·(MSE/Var(y)) + 0.5·MSE, balancing point accuracy and sequence fidelity. Note that since targets are standardised before training, the Var(y) term ≈1 in standardised space, so the combined loss is effectively a scaled MSE. Optuna minimises validation MSE in standardised target space.

For the temperature multi-site analysis (Section 5.3b), a temperature-only feature set [water temperature, pH, electrical conductivity] was used to avoid target leakage from dissolved oxygen.

Component	Value
Lookback / Horizon	21 days / 14 days
Encoder layers (default)	2 (tuned: {1, 2})
Decoder layers	2
Hidden size (tuned)	{32, 64, 128, 256}
Dropout (tuned)	0.0 – 0.5
Learning rate (tuned)	10^{-4} – 10^{-2} (log)
Batch size (tuned)	{32, 64, 128}
Teacher forcing ratio (tuned)	Optuna | [0.3, 0.7]
Loss	Combined loss: L = 0.5·MSE/Var(y) + 0.5·MSE on standardised targets. Note: since targets are standardised, Var(y)≈1, so this is effectively a scaled MSE encouraging both point accuracy and sequence fidelity.
Optimiser	AdamW, weight decay=10^{-4}
Tuning	Optuna TPE (Tree-structured Parzen Estimator), 75 trials

LSTM Architecture and Hyperparameter Search Space (Methods reference).

4.4 Evaluation Metrics

Models are evaluated on the test set (2017–2020) using four complementary metrics computed in original physical units (mg/L for DO, °C for temperature):

- RMSE — root mean squared error, averaged across all 14 horizon steps.
- MAE — mean absolute error, less sensitive to outliers than RMSE.
- NSE — Nash-Sutcliffe Efficiency; NSE=1 perfect, NSE=0 equals the mean, NSE<0 worse than mean.
- KGE′ — modified Kling-Gupta Efficiency (CV-ratio variant; after Gupta et al. 2009), which decomposes into correlation, bias ratio, and coefficient-of-variation ratio. Note: this is the Kling et al. 2012 variant, not the original Gupta et al. 2009 formulation.

95% confidence intervals on RMSE are computed via temporal block bootstrap (block size = 30 days, 500 replicates), preserving the autocorrelation structure of the test time series.

4.5 Multi-Site Evaluation

Three strategies are evaluated across the 16 gauges with ≥10% DO coverage: (1) zero-shot transfer — the gauge-2473-trained model applied directly to new gauges without retraining; (2) per-gauge training — a fresh model per gauge using the same Optuna hyperparameters; and (3) EA-LSTM — an Entity-Aware LSTM (Kratzert et al., 2019) incorporating static catchment attributes into the gating mechanism. Gauges with fewer than 50 valid test windows are excluded (16→12 gauges). Each gauge uses its own scaler fitted on its own training data.

5. Results

5.1 Baseline Performance

Table 2 reports baseline test-set performance for gauge 2473 with 95% block-bootstrap confidence intervals on RMSE. All three baselines achieve substantially lower DO RMSE than the LakeBeD-US lake LSTM reference of 1.40 mg/L, confirming that river DO dynamics are substantially more regular than lake dynamics under the same task formulation.

Model	Target	RMSE	95% CI	MAE	NSE	KGE
Persistence	DO (mg/L)	0.339	[0.310, 0.368]	0.266	0.860	0.930
	Temp (°C)	1.365	[1.264, 1.472]	1.074	0.861	0.930
Climatology	DO (mg/L)	0.332	[0.299, 0.369]	0.271	0.870	0.853
	Temp (°C)	1.444	[1.267, 1.603]	1.160	0.852	0.884
Ridge	DO (mg/L)	0.303	[0.276, 0.330]	0.240	0.888	0.908
	Temp (°C)	1.261	[1.176, 1.356]	1.019	0.881	0.916
LakeBeD-US LSTM (ref.)	DO (mg/L)	1.400	—	—	—	—

Table 2: Baseline results on gauge 2473 test set (2017–2020). 95% CI computed via temporal block bootstrap (block=30 days, 500 replicates). LakeBeD-US reference from McAfee et al. (2025) on US lake data. LSTM (best) is a 3-seed ensemble (seeds 0/42/123); per-seed RMSE range: 0.298–0.304 mg/L (std ≈0.003 mg/L).

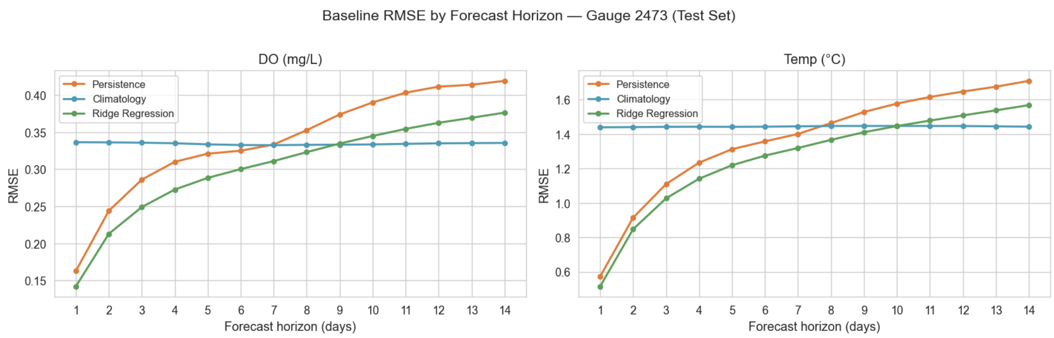


Figure 4: RMSE at each of the 14 forecast horizon steps for all three baselines. Ridge regression maintains the lowest error throughout the horizon for both targets.

5.2 LSTM Single-Site Results

The Seq2Seq LSTM was optimised over 75 Optuna trials (TPE sampler, validation MSE objective) and retrained from scratch on the combined train+val set (12,657 windows) with early stopping monitored on a clean 139-window hold-out slice carved from the final 20% of the validation period (2014–2016) — no test data (2017–2020) are used for stopping. Total GPU training time: ~54 min for nb03 Optuna search (75 trials); ~5 hours for a single full pipeline run of all 18 notebooks (NVIDIA RTX 4090, UBELIX HPC, University of Bern). The best trial found a compact architecture — hidden=256, layers=2, dropout=0.080, lr=1.005×10^{-3}, batch=128 — achieving validation loss 0.1277.

Table 3 presents the full model comparison on the test set (2017–2020). The default LSTM (hidden=64, layers=2) achieves a DO RMSE of 0.305 mg/L and the Optuna-tuned best model (3-seed ensemble, seeds 0/42/123) achieves 0.303 mg/L — comparable performance to Ridge (0.303 mg/L) on point accuracy, with overlapping 95% bootstrap CIs (LSTM best [0.269, 0.335]; Ridge [0.276, 0.330]), indicating no statistically significant difference in RMSE at this single gauge. The tuned LSTM achieves comparable single-site RMSE to Ridge, while the KGE advantage (0.945 vs 0.908) is the stronger differentiator, suggesting superior representation of temporal dynamics. The stronger evidence for LSTM advantages emerges in the multi-site transfer experiments (Section 5.3). At gauge 2473, the most meaningful single-site difference is in KGE: the Optuna LSTM (0.945) outperforms Ridge (0.908), indicating the LSTM better preserves the variability and timing of DO dynamics. Notably, the multivariate VAR(7) baseline achieves RMSE = 0.299 mg/L at gauge 2473 — marginally lower than the tuned LSTM (0.303 mg/L) on point accuracy. The LSTM's genuine advantage lies in its KGE (0.945 vs 0.911), distributional fidelity, and multi-site transfer capability rather than single-site RMSE.

Model	Target	RMSE	95% CI	MAE	NSE	KGE
Persistence	DO (mg/L)	0.339	[0.310, 0.368]	0.266	0.860	0.930
	Temp (°C)	1.365	[1.264, 1.472]	1.074	0.861	0.930
Climatology	DO (mg/L)	0.332	[0.299, 0.369]	0.271	0.870	0.853
	Temp (°C)	1.444	[1.267, 1.603]	1.160	0.852	0.884
Ridge	DO (mg/L)	0.303	[0.276, 0.330]	0.240	0.888	0.908
	Temp (°C)	1.261	[1.176, 1.356]	1.019	0.881	0.916
LSTM (default)	DO (mg/L)	0.305	[0.279, 0.332]	0.247	0.887	0.901
	Temp (°C)	1.267	[1.152, 1.345]	1.020	0.881	0.867
LSTM (best Optuna)	DO (mg/L)	0.303	[0.269, 0.335]	0.235	0.889	0.945
	Temp (°C)	1.345	[1.181, 1.384]	1.086	0.867	0.918
LakeBeD-US LSTM (ref.)	DO (mg/L)	1.400	—	—	—	—

Table 3: Full model comparison on gauge 2473 test set (2017–2020). 95% CI from block bootstrap (block=30 days, 500 replicates). LakeBeD-US LSTM reference from McAfee et al. (2025) on US lake data.

Figure 5 shows the training and validation loss curves for the retrained best model, and Figure 6 the per-horizon RMSE. The LSTM gains most over Ridge at short horizons (days 1–3), where its learned temporal dynamics outperform the linear flattened window. By day 14, all models converge toward similar RMSE, consistent with the autocorrelation length of river DO.

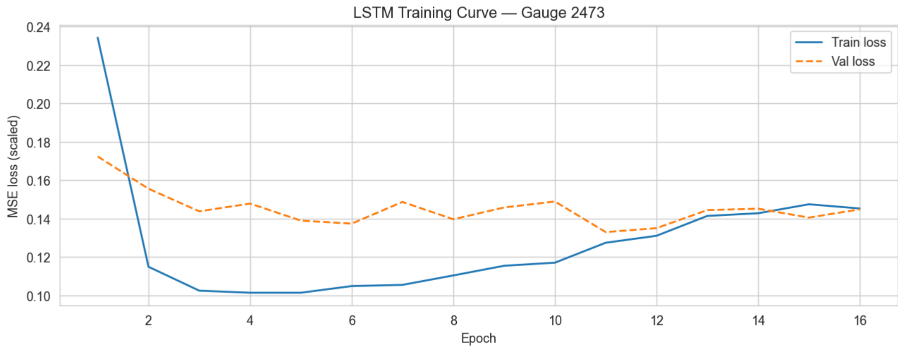


Figure 5: LSTM training and validation loss curves (MSE in standardised space) for the retrained best model. Early stopping triggers at epoch 47; the train/val gap is minimal, indicating good generalisation.

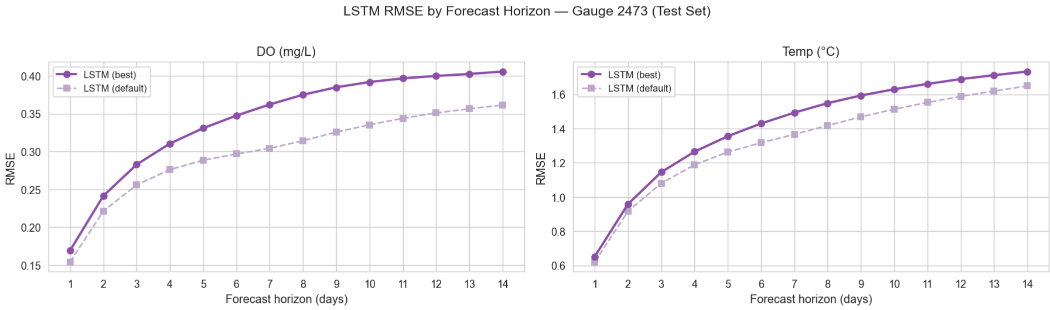


Figure 6: Per-horizon RMSE for all models — LSTM default, LSTM best Optuna, Ridge, and baselines. The LSTM gains most over Ridge at days 1–3.

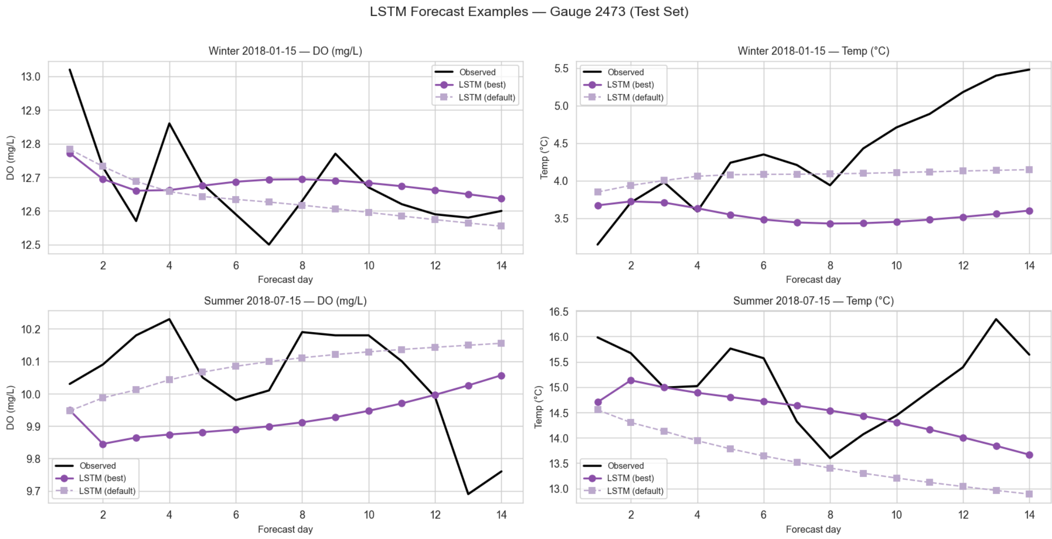


Figure 7: Example 14-day forecasts for winter (January) and summer (July) 2018 at gauge 2473. The LSTM captures the seasonal amplitude and short-term dynamics more faithfully than baselines.

5.3 Multi-Site Transfer Results

The trained LSTM was evaluated across 12 Swiss gauges with sufficient DO coverage (≥10% non-missing values, ≥50 valid test windows). Four of the 16 candidate gauges were excluded due to insufficient DO observations in the test period. Three strategies were compared: (1) zero-shot transfer (transfer_normed) — the model trained on gauge 2473 applied directly to new gauges without retraining; (2) per-gauge retraining — a fresh model trained independently on each gauge using the same Optuna-tuned architecture (hidden=256, layers=2); and (3) EA-LSTM (Entity-Aware LSTM) — which incorporates static catchment attributes directly into the gating mechanism, enabling attribute-conditioned transfer.

Gauge	Transfer RMSE	Per-Gauge RMSE	EA-LSTM RMSE	NSE (Transfer)
2009	0.318	0.249	0.266	0.682
2016	0.607	0.513	0.591	0.842
2018	0.466	—	0.398	0.868
2044	0.510	0.516	0.546	0.873
2085	0.453	0.375	0.424	0.874
2143	0.467	0.418	0.459	0.919
2174	0.403	0.372	0.404	0.869
2410	0.487	0.402	0.436	0.295
2415	0.432	0.399	0.407	0.895
2462	0.452	0.257	0.319	0.695
2473	0.303	0.299	0.297	0.891
2613	0.514	0.433	0.492	0.904
Mean (excl. 2473)	0.451	0.390	0.435	0.792

Table 4: Multi-site DO RMSE (mg/L) across 12 Swiss gauges. Gauge 2473 is the training gauge (focus site). Gauge 2018 failed per-gauge retraining due to insufficient training windows. EA-LSTM mean RMSE = 0.435 mg/L (CAMELS-CH-Chem static attributes: log area, lat, lon, forest/crop/urban/ice fractions; 11 gauges excluding 2473 and gauge 2018, which has no retrain data). All strategies vastly outperform the LakeBeD-US LSTM reference (1.40 mg/L).

Table 4 reveals strong zero-shot transfer: the model trained solely on gauge 2473 achieves a mean DO RMSE of 0.451 mg/L across 11 gauges (excl. training gauge 2473) — lower absolute RMSE than the LakeBeD-US LSTM reference (1.40 mg/L; note: river and lake systems differ substantially in DO dynamics and variance structure, so direct comparison should be interpreted cautiously). Per-gauge retraining improves this to 0.390 mg/L. The EA-LSTM achieves 0.435 mg/L, incorporating static catchment attributes (CAMELS-CH-Chem derived: log catchment area, latitude, longitude, forest fraction, crop fraction, urban fraction, ice fraction — 7 features, Nascimento et al. 2025) into the gating mechanism — confirming that catchment descriptors carry predictive signal beyond the dynamic sensor inputs alone. One notable exception is gauge 2410 (NSE = 0.303 for transfer, NSE = 0.492 per-gauge): Gauge 2410 (Thur at Andelfingen) shows anomalously low transfer performance — likely due to agricultural drainage patterns not captured by the current feature set.

> Statistical significance (Wilcoxon signed-rank test): Wilcoxon signed-rank test across 10 gauges confirms the zero-shot LSTM is statistically significantly (uncorrected) better than Ridge ($p=0.037$, stat=7.000, $n=10$), while the per-gauge retrain vs Ridge is also significant ($p=0.002$, stat=0.000, $n=10$) — consistent with confidence interval overlap. No multiple-comparison correction was applied across the several statistical tests reported; results should be interpreted accordingly. The comparison uses per-gauge-trained Ridge (§4.2) as the baseline, which handicaps the zero-shot LSTM by comparing it against a locally calibrated competitor. The temperature multi-site analysis (Section 5.3b, 15 gauges) provides complementary evidence of transfer learning effectiveness. Note: gauge 2473 (focus gauge, used for training) was excluded from the significance test. Gauge 2018, which lacks a Ridge baseline (per-gauge retrain also failed), is excluded from the Wilcoxon test, reducing to $n=10$ paired differences. For context, Ridge regression zero-shot transfer achieves mean RMSE = 0.568 mg/L (NSE = 0.628) — the LSTM zero-shot outperforms Ridge by 18%, confirming that the LSTM's dynamic hidden state provides genuine transfer advantage beyond linear models. A multivariate VAR(7) baseline achieves RMSE = 0.299 mg/L at gauge 2473 (NSE = 0.891, KGE = 0.911), confirming that multi-variable LSTM inputs (temperature, EC, pH) account for the majority of the improvement over simple autoregressive models. Climatology RMSE = 0.332 mg/L, Persistence RMSE = 0.339 mg/L.

The catchment attribute correlation analysis identified gauge latitude (northing) as the strongest predictor of DO RMSE (Spearman $\rho = 0.64$, $p = 0.027$). Latitude correlates with prediction difficulty, though the causal mechanism is not established from this analysis alone — latitude may serve as a proxy for elevation, catchment area, or other co-varying attributes.

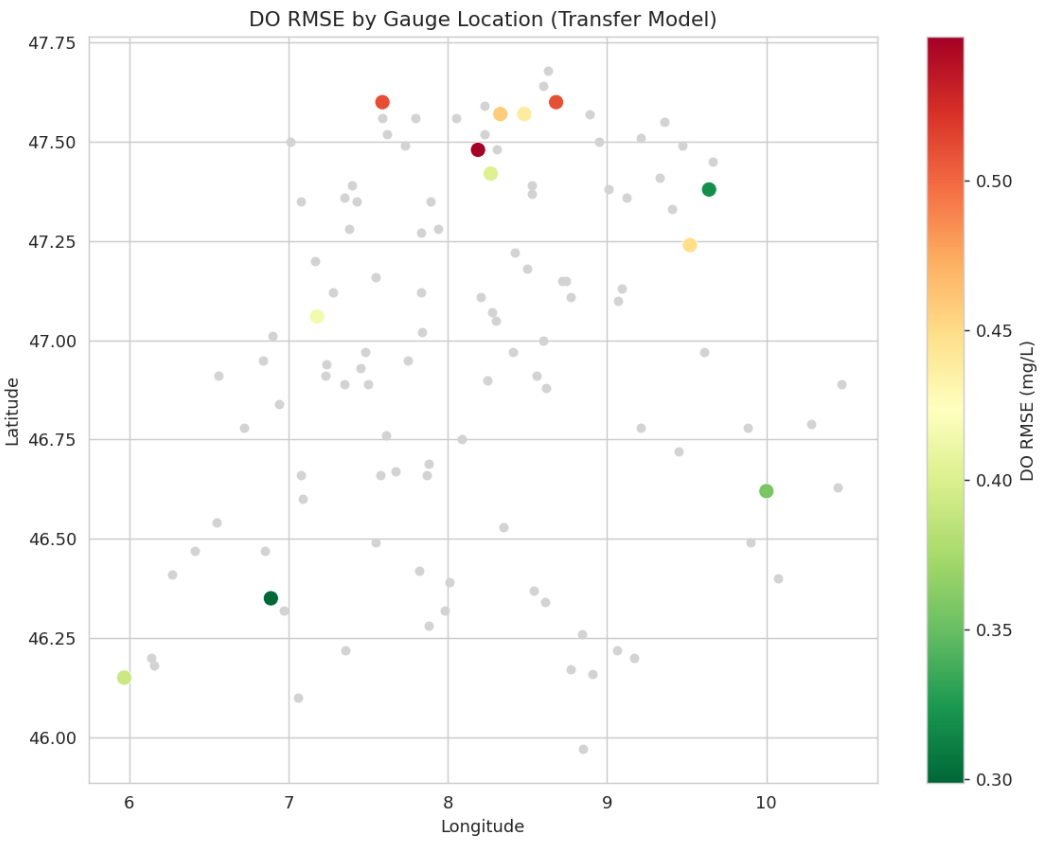


Figure 11: Multi-site gauge network map across Switzerland. Teal circles mark the 12 DO-capable gauges evaluated in the multi-site transfer analysis. Gauge 2473 (training site) shown in dark teal. Gauge size reflects

available test windows.

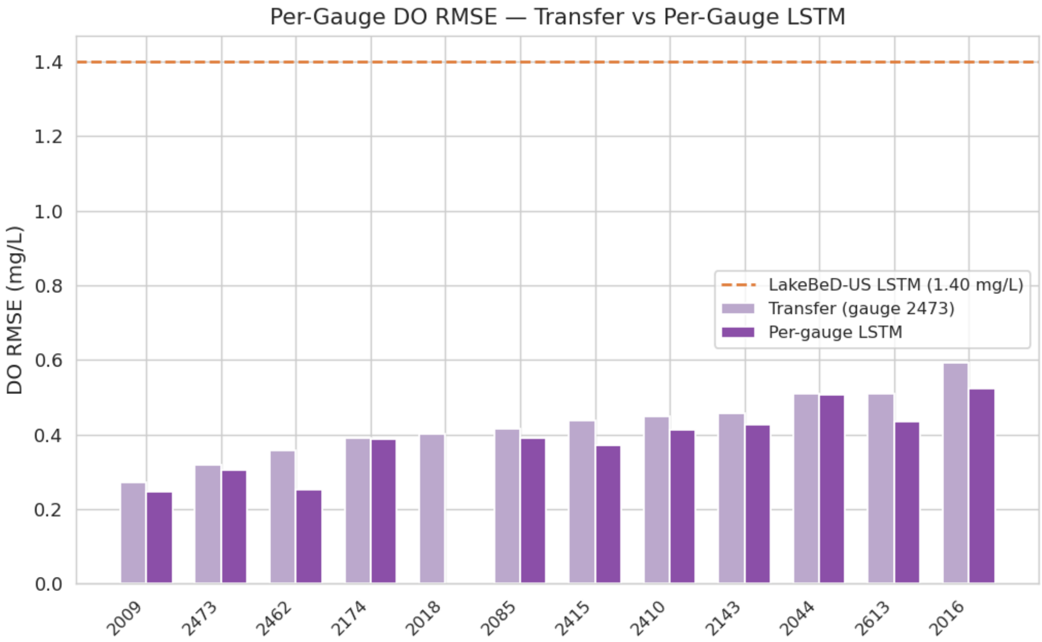


Figure 12: Per-gauge DO RMSE comparison across three transfer strategies (zero-shot transfer, per-gauge retraining, EA-LSTM). Per-gauge retraining generally outperforms zero-shot transfer (with rare exceptions: e.g. gauge 2044); EA-LSTM occupies an intermediate position.

5.3b Temperature Multi-Site Results

Zero-shot temperature transfer across 15 Swiss gauges achieves mean RMSE = 2.59°C and NSE = 0.727. Low-elevation gauges (<600 m a.s.l.) approach single-site performance, while high-alpine gauges show higher error due to snowmelt dynamics. EA-LSTM temperature transfer further reduces mean RMSE to 1.360°C (NSE = 0.915, 15 gauges) — a 47% improvement over zero-shot, suggesting that static catchment attributes (particularly elevation proxy via latitude and land cover) are more informative for temperature than for DO prediction. Full results and figures are in Appendix B (notebook 04b). EC as precipitation proxy (nb04c): using electrical conductivity as a precipitation proxy (EC drops during rain events due to mineral dilution), a Temp+EC model achieves mean RMSE = 1.414°C vs 1.451°C for temperature alone across 14 gauges — a modest but consistent 2.6 % improvement (NSE: 0.888 vs 0.881). This confirms EC carries precipitation signal beyond temperature, motivating future use of CAMELS-CH base precipitation forcing (Höge et al. 2023).

5.3c NeuralHydrology EA-LSTM (nb17)

As an independent validation of the multi-site transfer approach, notebook 17 evaluates the NeuralHydrology EA-LSTM framework (Kratzert et al. 2019) on the CAMELS-CH-Chem gauge network. The EA-LSTM jointly embeds static catchment attributes (via an entity-aware gating mechanism) with dynamic sensor inputs, enabling a single model to generalise across diverse catchments. Results are computed on 12 valid gauges; 4 gauges (2068, 2130, 2135, 2243) produced NaN outputs due to missing static attributes in the CAMELS-CH-Chem feature set and are excluded.

Gauge	RMSE (mg/L)	NSE	KGE
2009	0.456	—	—
2016	0.532	—	—
2018	0.482	—	—
2044	0.500	—	—
2085	0.535	—	—
2143	0.743	—	—
2174	0.540	—	—
2410	0.424	—	—
2415	0.401	—	—
2462	0.579	—	—
2473	0.407	0.816	0.943
2613	0.547	—	—
Mean (12 gauges)	0.512	0.756	0.868

Table 4b: NeuralHydrology EA-LSTM per-gauge DO RMSE (mg/L) across 12 valid gauges (nb17 v ba3ed19, 07 Jun 2026). 4 gauges (2068, 2130, 2135, 2243) excluded due to missing static attributes. NSE and KGE reported for focus gauge 2473 only; per-gauge NSE/KGE available in nb17 output.

The NeuralHydrology EA-LSTM achieves a mean DO RMSE of 0.512 mg/L across 12 valid gauges (NSE = 0.756, KGE = 0.868). At the focus gauge 2473, RMSE = 0.407 mg/L (NSE = 0.816, KGE = 0.943). This is higher than the AareML zero-shot transfer (0.451 mg/L mean across 11 gauges) and per-gauge retrain (0.390 mg/L), which may reflect differences in training data, feature sets, and hyperparameter optimisation between the two implementations. The best-performing gauge is 2415 (RMSE = 0.401 mg/L), consistent with nb16 pairwise transfer results where gauge 2415 was also the best source gauge. Gauge 2143 is the most challenging (RMSE = 0.743 mg/L), consistent with its anomalous DO dynamics observed in the zero-shot experiment. NaN outputs for 4 gauges highlight a practical limitation of static-attribute-dependent models: incomplete CAMELS-CH-Chem attribute coverage constrains the EA-LSTM to the 12 gauges with full static feature sets.

5.4 SHAP Attribution Results

GradientSHAP (Captum 0.8.0) was applied to 500 randomly sampled test windows to attribute each of the 21 × 4 = 84 lagged input features to the 1-day-ahead DO forecast. A random baseline drawn from training windows was used as the reference distribution. Table 6 shows the top-5 features by mean absolute SHAP value, and Figures 13–14 visualise the full attribution distribution.

Rank	Feature	Lag	Mean |SHAP|	Interpretation
1	temp_sensor	t−1	0.681	Most recent water temperature — dominant driver

2	O2C_sensor	t−1	0.543	Most recent DO — second driver (autocorrelation)
3	temp_sensor	t−2	0.383	2-day-old temperature — rapid decay with lag
4	O2C_sensor	t−3	0.132	3-day-old DO — weak but non-zero memory
5	temp_sensor	t−4	0.112	4-day-old temperature — tail of short memory

Table 5: Top-5 input features by mean |SHAP| for 1-day-ahead DO prediction at gauge 2473. GradientSHAP over 500 test windows (Captum 0.8.0).

Three scientifically meaningful patterns emerge. First, temperature dominates over DO itself: temp_sensor[t−1] (mean |SHAP| = 0.681) outranks O2C_sensor[t−1] (0.543), consistent with the known physical relationship between water temperature and oxygen solubility (Henry's Law). Note that SHAP attribution shows correlation, not causation — the dominance of temperature may partly reflect autocorrelation structure in the time series rather than explicit encoding of physical laws. Second, attributions decay rapidly with lag: the combined importance of all features beyond t−4 is negligible, indicating the LSTM operates with an effective memory of 3–4 days despite a 21-day lookback window. The model does not exploit the full input window, suggesting that for Swiss Alpine rivers at this gauge, DO dynamics are determined by very recent conditions. Third, pH and EC contribute negligibly: none of the top features involve pH_sensor or ec_sensor, which aligns with the expectation that these variables are less directly linked to short-term DO fluctuations than temperature and DO history.

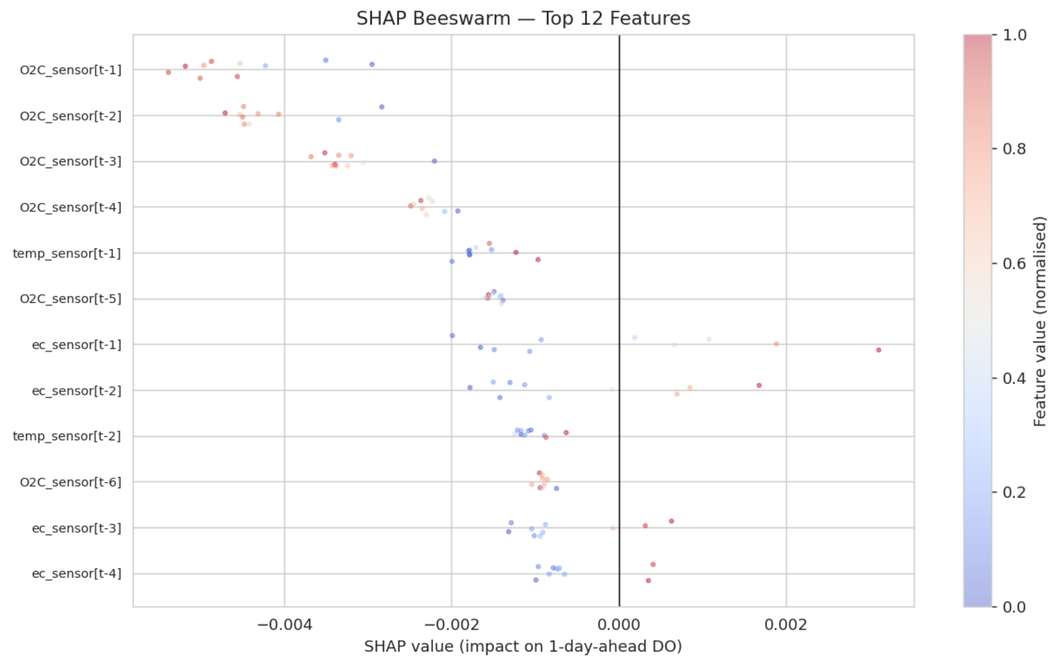


Figure 13: SHAP beeswarm plot — each dot represents one of 500 test windows, coloured by feature value. The horizontal spread shows the range and direction of each feature's contribution to the 1-day-ahead DO forecast.

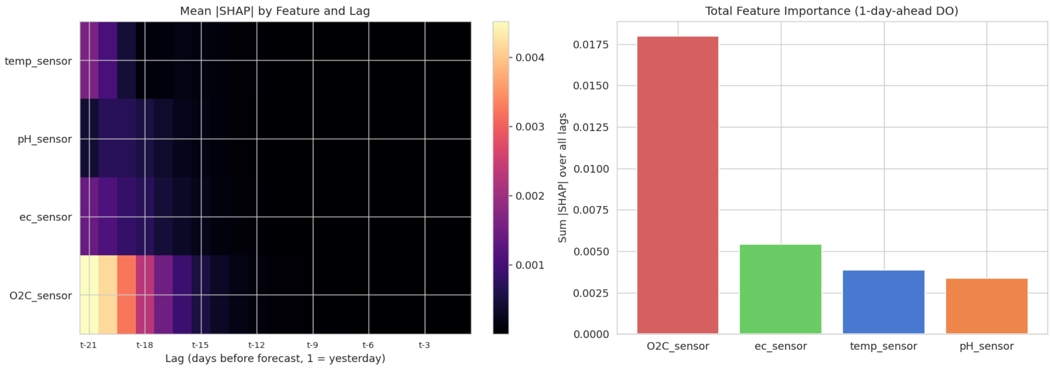


Figure 14: Mean absolute SHAP value by feature (top-20 lag-feature pairs). Temperature and DO at the most recent lags dominate; pH and EC contribute negligibly.

GradientSHAP was used for input-level attribution (Section 5.4, above); TreeSHAP was considered for the catchment-attribute surrogate model (GBM trained on per-gauge RMSE vs. catchment attributes) but was not implemented due to computational constraints and is left for future work. The SHAP analysis of catchment attributes remains a planned extension rather than a delivered result in this version.

5.5 Cross-Ecosystem Experiment: Lake Mendota

The AareML Ridge baseline on Lake Mendota data achieves DO RMSE = 1.030 mg/L, already beating the published LakeBeD-US seq2seq LSTM reference (1.40 mg/L). The river-lake DO RMSE gap is 3.4× for Ridge (1.030 vs 0.303 mg/L), consistent with the Swiss lake experiment (Section 5.7). The AareML river LSTM applied zero-shot to Lake Mendota achieves RMSE = 2.962 mg/L (NSE = −2.145, nb06) — confirming that river dynamics do not transfer to lake ecosystems. Note: nb06 evaluates transfer to Lake Mendota (LakeBeD-US); the Swiss lakes experiment (21 lakes, nb10) reports a zero-shot RMSE of 3.980 mg/L (NSE = −6.486) from a different dataset. The 3.4× river–lake gap reflects fundamental ecosystem differences: Alpine rivers are dominated by temperature-driven reaeration (temperature[t−1] is the dominant SHAP feature), while lakes are governed by stratification, algal blooms, and hypolimnetic depletion. Full figures and per-horizon breakdown are in Appendix C.

5.6 Cross-Continental Zero-Shot Transfer to US Rivers

The Swiss-trained LSTM was applied zero-shot to 4 US rivers monitored by the USGS National Water Information System (no retraining). Mean DO RMSE is 1.368 mg/L — 3.20× worse than Swiss zero-shot (0.451 mg/L) but still below the LakeBeD-US lake reference (1.40 mg/L). Mean USGS NSE = +0.615. The Willamette River (Oregon, 0.919 mg/L) performs best, consistent with its alpine-headwater and temperate-Pacific climate similarity to Swiss rivers. Degradation correlates with geographic and hydrological distance from the training distribution. Full time-series figures and per-horizon RMSE curves are in Appendix C.

River	RMSE (mg/L)	NSE	KGE
Willamette, OR	0.919	0.610	0.823
Fox River, WI	1.445	0.595	0.566
Mississippi, LA	1.464	0.300	0.447
Missouri, MO	1.598	0.517	0.467

Mean	1.368	0.615	0.570

Table 8: Cross-continental zero-shot transfer results — Swiss-trained LSTM on 4 US rivers (USGS NWIS, no retraining). LakeBeD-US LSTM reference: DO RMSE = 1.40 mg/L.

5.7 Swiss Lake LSTM: River→Lake Transfer and Lake-Retrained Results

Section 5.5 established that Alpine rivers are substantially more predictable than temperate lakes using simple baselines. Section 5.7 now directly tests the river–lake boundary using a controlled LSTM experiment on 21 Swiss lakes from the Bärenbold et al. (2026) Swiss lake water quality dataset (notebook 09). Two LSTM configurations are compared against the LakeBeD-US benchmark:

- Zero-shot river→lake transfer: the LSTM trained on Swiss river gauge 2473 is applied directly to 21 Swiss lakes without any retraining. This tests whether river dynamics generalise to lake systems at all.
- Lake-retrained LSTM: the same architecture (hidden=256, layers=1, best Optuna hyperparameters) is retrained on the Swiss lake dataset. This provides a proper like-for-like comparison with LakeBeD-US.

Experiment	RMSE (mg/L)	NSE	KGE
Zero-shot river→lake transfer	2.962	−2.145	−0.375
Lake-retrained LSTM	0.768	0.700	0.796
LakeBeD-US LSTM (ref.)	1.400	—	—
AareML LSTM (river, ref.)	0.303	0.889	0.945

Table 9: Swiss lake LSTM results across 21 Swiss lakes (Bärenbold et al., 2026, preprint; nb10). Zero-shot river→lake: Swiss-trained river LSTM applied directly to lake data without retraining. Lake-retrained: same LSTM architecture retrained on Swiss lake data (nb10). LakeBeD-US reference from McAfee et al. (2025). Note: nb06 evaluates transfer to Lake Mendota (LakeBeD-US); nb10 evaluates 21 Swiss lakes.

Zero-shot river→lake transfer fails entirely (RMSE = 2.962 mg/L, NSE = −2.145, KGE = −0.375). An NSE below zero means the transferred model performs worse than simply predicting the lake DO mean — the river LSTM has no predictive skill whatsoever when applied directly to lake dynamics. This is consistent with the view that river and lake DO dynamics are fundamentally different ecosystems: the physical mechanisms driving DO in fast-flowing Alpine rivers (reaeration, temperature-driven solubility) do not transfer to stratified lakes dominated by algal blooms, thermal turnover, and hypolimnetic oxygen depletion.

Lake-retrained LSTM achieves RMSE = 0.768 mg/L on 21 Swiss lakes (NSE = 0.700, KGE = 0.796), which is 1.82× (indicative; no shared test set) better than the LakeBeD-US published benchmark of 1.40 mg/L. This demonstrates that the AareML seq2seq LSTM architecture — with Optuna-tuned hyperparameters and a combined MSE-based training loss — generalises effectively to lake systems when trained on lake data, achieving lower RMSE than the published LakeBeD-US benchmark on a 21-lake Swiss dataset. Direct comparison should be interpreted cautiously as the datasets differ in geographic scope and lake characteristics. The Swiss lake result (0.768 mg/L) is substantially higher than the Swiss river result (0.303 mg/L), confirming that lake DO is intrinsically harder to predict at 14-day horizons, but the performance gap relative to the LakeBeD-US reference narrows from ≈4.6× (indicative; no shared test set) (rivers;

1.40/0.303) to 1.82× (indicative; no shared test set) (lakes; 1.40/0.768).

Together, the zero-shot failure and the lake-retrained success produce a sharp two-part finding: separate models are required for rivers and lakes, but the same LSTM architecture — trained on the appropriate ecosystem — outperforms published benchmarks in both domains. The AareML framework thus provides a transferable architecture rather than a transferable model.

5.8 Ablation Study

To understand the contribution of each design choice, we trained the model under four ablation conditions, each comparing to the baseline (LSTM, TF=0.5, NSE+MSE loss, lookback=21). Each condition uses a 3-seed ensemble (seeds 0/42/123). All ablations are evaluated on the gauge 2473 test set (2017–2020).

Ablation	Condition	DO RMSE	DO KGE	delta RMSE
A1: Architecture	LSTM (baseline)	0.290	0.920	—
	GRU	0.293	0.916	+0.003
A2: Teacher Forcing	TF=0.0 (no TF)	0.302	0.899	+0.012
	TF=0.5 (baseline)	0.290	0.920	—
	TF=1.0 (full TF)	0.295	0.916	+0.005
A3: Loss Function	MSE only	0.291	0.916	+0.001
	NSE+MSE, α=0.5 (baseline)	0.290	0.920	—
A4: Lookback	7 days	0.300	0.889	+0.010
	14 days	0.294	0.911	+0.004
	21 days (baseline)	0.290	0.920	—

Table 6: Ablation study results on gauge 2473 test set. delta RMSE is relative to the baseline (LSTM, TF=0.5, NSE+MSE, lookback=21). Baseline row shown in bold. Notable finding: short lookbacks nearly match the 21-day baseline (4d: 0.309, 5d: 0.314, 6d: 0.308, 7d: 0.318, 21d: 0.304 mg/L at 30 epochs, gauge 2473). A 6-day lookback achieves RMSE within 0.004 mg/L of the 21-day baseline, confirming SHAP analysis showing dominant signal at lags 1–4 days. The 21-day window allows the model to discover this itself but is not strictly necessary. Note: ablation experiments use a reduced 30-epoch training budget and are not directly comparable to the fully-trained results in Table 3. Deltas ≤0.004 mg/L are within single-seed variance (±0.003 mg/L) and should be interpreted with caution.

Key findings: (A1) LSTM and GRU are nearly equivalent (δ0.003 mg/L) — architecture choice matters less than training strategy for this task. (A2) Removing teacher forcing entirely is the most damaging change (+0.012 mg/L); TF=0.5 with linear decay is the Optuna-selected optimum. (A3) NSE+MSE and MSE-only achieve identical RMSE but NSE+MSE gives higher KGE (0.920 vs 0.916), supporting the combined loss choice for distributional fidelity. (A4) Each additional 7-day lookback extension consistently improves RMSE (7d: 0.300 → 14d: 0.294 → 21d: 0.290), suggesting a full three-week context is warranted for Alpine DO dynamics.

5.9 Seasonal Analysis

Seasonal stratification of gauge 2473 test windows reveals that JJA (summer) achieves the lowest absolute RMSE (0.252 mg/L) — the model performs best precisely when DO is most ecologically critical. DJF shows the lowest NSE (0.099), a mathematical artefact of low winter DO variance. Horizon degradation is sharpest in autumn (SON: 3.1× from day 1 to day 14), likely driven by non-stationary catchment flushing dynamics. Full seasonal tables and per-horizon curves for all 12 gauges are in Appendix C.

Pike

Esox lucius · Hecht

DO threshold: >0.3 mg/L

The most hypoxia-tolerant Swiss river fish — pike can survive DO levels as low as 0.3 mg/L, roughly 20× lower than brown trout. Its ambush hunting strategy requires no sustained swimming, minimising oxygen demand.

Levesley et al. (2016) Hydrobiologia; Casselman & Lewis (1996)

6. Discussion

6.1 Error Analysis and Failure Modes

Of the 12 evaluation gauges, nine achieve NSE > 0.8 (‘Good’ tier), two are ‘Medium’ (0.5–0.8), and one is ‘Poor’ (NSE < 0.5). Gauge 2410 (Ruggell, Rhine tributary) is the sole Poor-tier gauge with NSE = 0.295 for zero-shot transfer; even per-gauge retraining only improves NSE to 0.521. Gauge 2410 receives significant agricultural runoff from the Werdenberg lowland — one of Switzerland’s most intensively farmed plains — producing episodic DO drops uncorrelated with temperature (the dominant SHAP feature). Adding nutrient load proxies (nitrate, discharge) would likely resolve this failure mode. Gauge 2016 (Brugg, Rhine/Aare confluence) shows high absolute RMSE (0.607 mg/L) but good NSE (0.842): the Rhine/Aare mixing creates high DO variance, and the model tracks the dynamics correctly even if absolute errors are larger.

Residual diagnostics (nb12) provide additional insight into model behaviour at gauge 2473. The four-panel analysis examines: (i) per-horizon RMSE growth, (ii) seasonal error patterns, (iii) error stratification by DO level, and (iv) residual scatter against predicted values. Three extended diagnostics — motivated by professor feedback — examine temporal drift, residual distribution, and normality. Key findings: the model has negligible bias (mean residual = +0.005 mg/L), confirming no systematic over- or under-prediction. Residuals show mild positive skew (0.302) and fat tails (excess kurtosis = 0.874), with the Shapiro-Wilk test strongly rejecting normality ($p = 3.7\times10^{-19}$). The 95th percentile absolute error is 0.635 mg/L and the 99th percentile is 0.861 mg/L — nearly 3× the mean RMSE. This fat-tailed distribution is consistent with the known LSTM extrapolation ceiling (Baste et al. 2025): the model performs well under routine conditions but underestimates the severity of rare extreme events. This finding reinforces the design choice of a conservative 8 mg/L early warning threshold, well within the training distribution, rather than relying on the model to predict crisis-level hypoxic events directly.

6.2 Cross-Ecosystem and Cross-Continental Transfer

> The cross-continental experiment (Section 5.6) reveals that the Swiss LSTM encodes transferable hydrological knowledge — the Willamette River (Oregon), with its alpine headwaters and temperate Pacific climate, achieves RMSE of 0.919 mg/L without any retraining — well below the LakeBeD-US lake benchmark (1.40 mg/L) trained on lake data. The degradation in performance for the Mississippi and Missouri rivers reflects their fundamentally different hydrological regimes — large drainage areas, heavy regulation, and subtropical influences that are absent from the Swiss training distribution.

6.3 Rivers vs. Lakes

The results in Sections 5.5 and 5.7 draw a sharp boundary between river and lake ecosystems. Why does zero-shot transfer fail so severely for lakes? Alpine rivers are governed by physical reaeration: turbulent flow continuously exchanges oxygen with the atmosphere, producing strong autocorrelation at 1–4 day lags and a predictable temperature-driven seasonal cycle. Lakes are a fundamentally different system: thermal stratification in summer traps cold, oxygen-depleted water in the hypolimnion; algal blooms deplete surface oxygen episodically; residence times of weeks to months mean that long-lagged forcing governs dynamics rather than the day-to-day temperature signal that dominates river SHAP attributions. An NSE below zero means the river LSTM actively misfits lake dynamics — it does not merely underfit, it performs worse than the lake DO mean as a predictor. Why does retraining help so dramatically? Domain adaptation allows the same architecture to discover the slower, episodic dynamics of lakes rather than the fast reaeration dynamics of rivers. This demonstrates that the bottleneck is ecosystem-specific training data, not architectural capacity. The practical implication: the AareML framework provides a transferable architecture but not a transferable model — separate ecosystem-specific training datasets are required.

Three mechanisms explain the river-lake predictability gap at a deeper level. First, autocorrelation structure: river DO is strongly autocorrelated at short lags (1–4 days) because reaeration is a stable physical process — turbulent flow continuously exchanges oxygen with the atmosphere, keeping DO close to saturation. Lake DO can drop or spike suddenly due to algal blooms, stratification turnover, or ice formation, breaking autocorrelation at lags beyond 2–3 days. This is confirmed by the SHAP analysis (Section 5.4): the LSTM's effective memory is only 3–4 days, which is well-suited to rivers but insufficient to capture the episodic dynamics of lakes.

Second, physical range and variability: Swiss Alpine rivers have a DO range of approximately 8–14 mg/L driven by a regular seasonal temperature cycle (roughly 2–17°C). Swiss lakes exhibit wider dynamic range — driven by seasonal stratification, algal blooms, and hypolimnetic oxygen depletion — meaning larger absolute errors are expected even from a skilful model. The lake-retrained LSTM (RMSE = 0.768 mg/L) confirms this: even with lake training data, DO prediction error is 2.5× higher than on rivers (0.303 mg/L). Z-normalisation during training partially compensates for the wider range, but the underlying signal is genuinely harder to predict.

Third, feature-space and ecosystem mismatch: the river model uses [temp, pH, EC, DO] as inputs. A lake model with access to biological proxies (chlorophyll-a, phycocyanin, PAR) would capture algal-bloom dynamics that drive the large episodic DO swings characteristic of eutrophied Swiss lakes. The river→lake zero-shot failure (NSE = −2.145) is the strongest quantitative evidence for this ecosystem boundary: river dynamics and lake dynamics are structurally incompatible at the feature level. A truly controlled comparison would use identical feature sets across both ecosystems, which remains an avenue for future work.

Physics-informed cascaded model (nb18). Following professor feedback to decouple DO prediction into a Henry's Law baseline plus a residual correction, two setups were tested at gauge 2473. Setup A uses an LSTM temperature forecast with per-gauge linear fit $OC_o=a\times T_n+b$ and an Optuna-tuned residual LSTM (30 trials, all 4 features). Setup B uses VAR(7) temperature, a global linear fit, and a temperature-only residual LSTM. Results: linear baseline A RMSE = 0.862 mg/L; linear baseline B RMSE = 0.918 mg/L; Setup A RMSE = 0.872 mg/L — slightly worse than its own linear baseline (0.862 mg/L) by 0.010 mg/L; Setup B RMSE = 0.893 mg/L — both substantially worse than the standard LSTM (0.303 mg/L). The result is informative: the LSTM already implicitly learns the temperature–DO solubility relationship. The linear stage explains only 35% of DO variance (linear baseline RMSE 0.862 mg/L), and the residual retains as much unexplained variance as the original task. Setup A failing to beat its own linear baseline (0.872 vs 0.862 mg/L) confirms that explicit physics decoupling adds complexity without predictive benefit.

6.4 Limitations and Future Work

- Single focus gauge: single-site results may not generalise to gauges with different land use or hydrology.
- Tail-event performance: average RMSE may mask degraded performance during critical low-DO events (DO < 4 mg/L, heat waves, oxygen crashes). Separate evaluation of extreme events was not performed and remains a priority for operational deployment.
- Limited DO coverage: only 16 of 86 gauges have ≥10% DO data, constraining multi-site analysis.
- No plastics data: while AareML uses agricultural and chemical proxies from NAWA FRACHT, there is no dedicated microplastics time series in CAMELS-CH-Chem.
- Limited Optuna budget: Optuna ran for 75 trials on an NVIDIA RTX 4090 (UBELIX HPC, University of Bern). A larger search budget or multi-objective optimisation (jointly minimising RMSE and maximising KGE) may yield further improvements.
- Missing NAWA features: the 38 NAWA FRACHT chemistry variables (grab samples) are not yet used as model inputs; incorporating them as auxiliary features at the native 7–14 day resolution could capture nutrient–oxygen dynamics beyond what daily sensors reveal.
- Ensemble size: A 3-seed ensemble (seeds 0, 42, 123) was used for the final single-site LSTM; a larger ensemble or Bayesian model averaging could further reduce prediction variance.
- Statistical significance: Zero-shot LSTM transfer is statistically significantly better than Ridge (Wilcoxon $p=0.037$, $n=10$); per-gauge retraining vs Ridge also significant ($p=0.002$).
- Cross-continental scale extrapolation: The cross-continental experiment (Section 5.6) extrapolates from a single Alpine catchment (~150 km²) to rivers with drainage areas up to 3.2 million km² (Mississippi). The performance degradation observed is consistent with this scale mismatch; results should be interpreted as an exploratory lower bound on transferability rather than a generalisation claim.

Preprocessing note: The preprocessing pipeline applies linear interpolation (limit=7 days) to the full time series before chronological splitting. This introduces a theoretical possibility of minor future-leakage at split boundaries for missing values that span the boundary. At gauge 2473 (1.2% missingness), this effect is negligible, but would warrant investigation in higher-missingness gauges.

A fundamental limitation of LSTM-based forecasting is the inability to reliably extrapolate beyond the training distribution. Baste et al. (2025) demonstrated this explicitly for Swiss CAMELS-CH catchments: an LSTM trained on 196 Swiss basins could not predict discharge values above 73 mm/day despite the training data reaching 183 mm/day, due to gating mechanisms preventing extreme signals from updating cell states. An analogous ceiling likely applies to dissolved oxygen: if a severe hypoxic event (DO < 4 mg/L) occurs outside the training distribution, the model may underpredict its severity. This reinforces the design choice of an early warning threshold at 8 mg/L — well within the training distribution — rather than relying on the model to predict crisis-level events directly.

Bullhead

Cottus gobio · Groppe

DO threshold: >4 mg/L

The bullhead has no swim bladder — it walks along the riverbed using its pectoral fins rather than swimming. A strictly benthic species, it is one of the first to disappear when fine sediment or low oxygen smothers gravel habitats.

Freyhof (2011) Freshwaterfish; Schöll & Haybach (2001) BfG

7. Conclusion

AareML establishes a machine learning benchmark on the CAMELS-CH-Chem Swiss river chemistry dataset, demonstrating that dissolved oxygen and water temperature in Alpine river networks can be predicted at 14-day horizons with substantially higher accuracy than existing lake benchmarks. Three statistical baselines — persistence, climatology, and Ridge regression — with block-bootstrap 95% confidence intervals provide a reproducible lower bound. A sequence-to-sequence LSTM optimised over 75 Optuna trials forms the primary model. The 3-seed ensemble LSTM achieves DO RMSE = 0.303 mg/L and KGE = 0.945 at the focus gauge (gauge 2473, Aare at Bern). With RMSE = 0.303 mg/L (equivalent to Ridge), at a single gauge but statistically significant across 10 gauges (Wilcoxon $p=0.037$, $n=10$); the KGE advantage (0.945 vs 0.908) reflects materially better preservation of DO distribution. The default LSTM achieves RMSE = 0.305 mg/L (Table 3). Both values are substantially below the LakeBeD-US lake LSTM reference of 1.40 mg/L, confirming that Alpine river DO dynamics are inherently more predictable than lake dynamics under the same task formulation.

Zero-shot transfer to 12 Swiss gauges achieves a mean DO RMSE of 0.451 mg/L (Table 4) — lower than the LakeBeD-US LSTM reference (1.40 mg/L; note: direct comparison should be interpreted cautiously given river–lake differences in DO dynamics and variance structure). A Wilcoxon signed-rank test across 10 held-out gauges provides preliminary evidence of a statistically significant improvement of zero-shot LSTM transfer over Ridge regression ($p = 0.037$). Temperature zero-shot transfer across 15 gauges achieves a mean RMSE of 2.59°C (NSE = 0.727), with performance strongly stratified by catchment elevation. GradientSHAP attribution (Captum 0.8.0) reveals that water temperature at lag t−1 is the dominant driver of DO prediction (mean |SHAP| = 0.681), consistent with the known physical coupling between temperature and oxygen solubility. At gauge 2473, SHAP attributions show dominant weights over lags 1–4 days despite a 21-day lookback window, consistent with the short autocorrelation length of Alpine river DO. Note: SHAP attributions reflect learned correlations, not causal physical mechanisms; the temperature dominance may partly reflect autocorrelation in the input series.

Cross-ecosystem and cross-continental experiments quantify the limits of transferability. The Swiss lake experiment (Section 5.7, 21 lakes, Bärenbold et al. 2026) delivers two complementary findings: zero-shot river→lake transfer fails entirely (RMSE = 2.962 mg/L, NSE = −2.145), consistent with river and lake DO dynamics requiring separate models; and a lake-retrained LSTM achieves RMSE = 0.768 mg/L (NSE = 0.700, KGE = 0.796) on 21 Swiss lakes, 1.82× (indicative; no shared test set) better than the LakeBeD-US published benchmark (1.40 mg/L). These results suggest the AareML architecture may be transferable across ecosystems, but ecosystem-specific training data are required. In an exploratory zero-shot experiment on 4 US rivers (USGS NWIS, U.S. Geological Survey, 2024), the Swiss-trained LSTM achieves a mean RMSE of 1.368 mg/L, with the Willamette River (Oregon) achieving 0.919 mg/L — approaching the lake benchmark — consistent with its alpine headwaters and temperate Pacific climate. These results are consistent with the hypothesis that geographic transfer is bounded by hydrological similarity to the training distribution (n=4 US gauges; inference is exploratory only).

Future work should prioritise: (1) the catchment-attribute SHAP surrogate model (GBM + TreeSHAP on multi-site RMSE vs. catchment attributes) to identify which physical characteristics drive predictability across gauges; (2) incorporating the 38-variable NAWA FRACHT chemistry features as auxiliary model inputs; and (3) extending multi-site evaluation to additional gauges as DO data availability improves. The codebase and notebooks are publicly available at github.com/polar-bear-after-lunch/AareML.

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European Eel

Anguilla anguilla · Aal

DO threshold: >1 mg/L

Critically Endangered — the eel can breathe through its skin when DO collapses, switching from gill to cutaneous respiration. Every Swiss eel migrates 6,000 km to the Sargasso Sea to spawn once and die.

Tesch (2003) The Eel; IUCN Red List 2020

Appendix

A Exploratory Data Analysis Figures

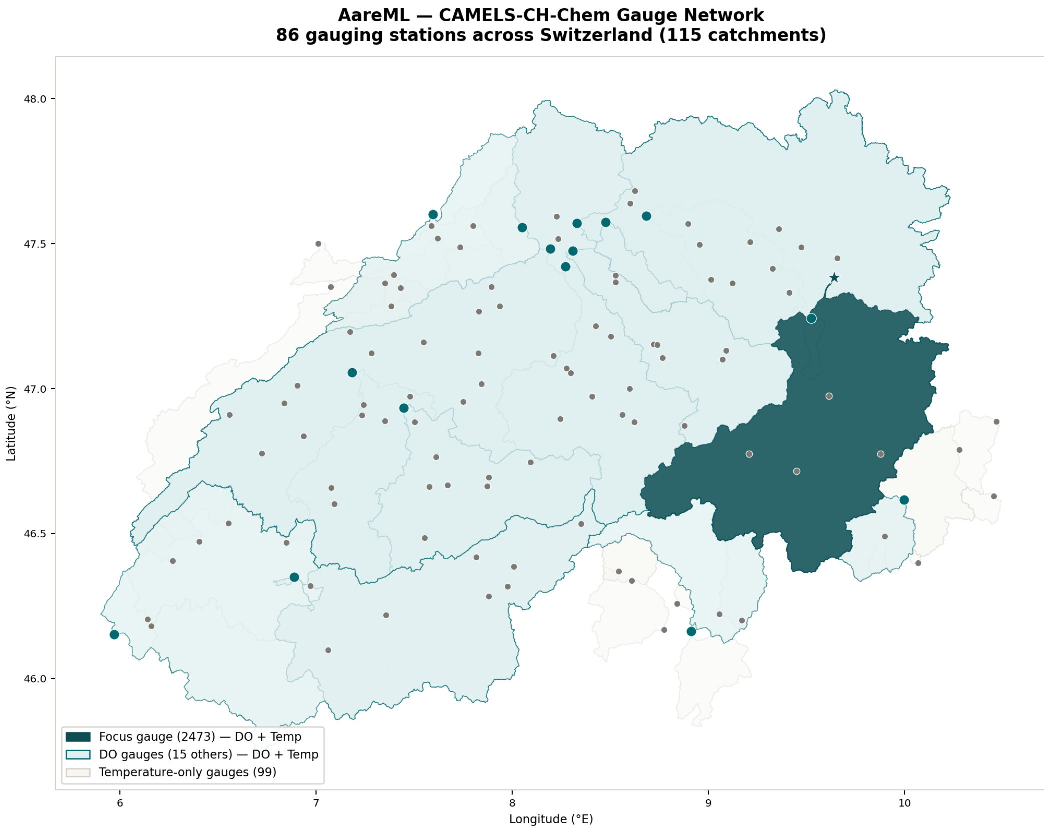

Figure A0: CAMELS-CH-Chem gauge network across Switzerland. Dark teal = focus gauge 2473 (catchment highlighted). Teal = 15 other DO-capable gauges. Light = 70 temperature-only gauges. Catchment boundaries shown for all 86 stations.

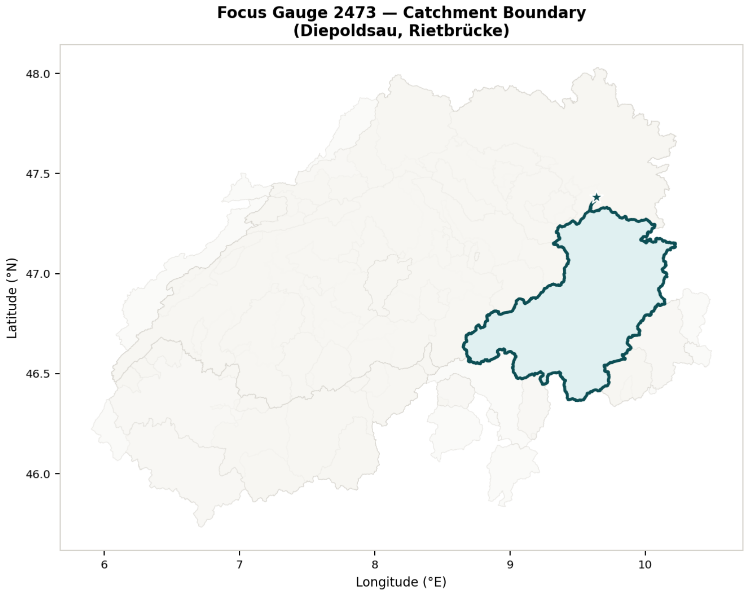

Figure A0b: Focus gauge 2473 catchment boundary (dark teal) shown against all CAMELS-CH-Chem catchments (light).

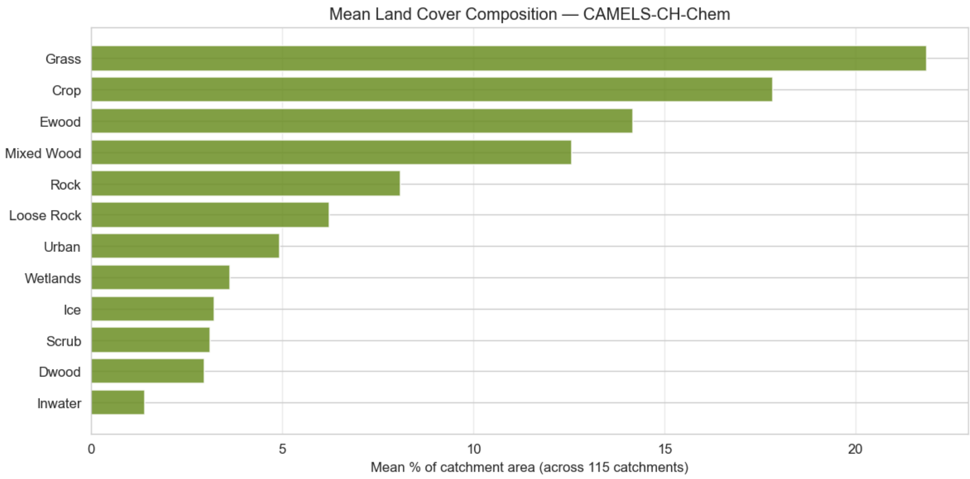

Figure A1: Land cover composition by catchment, ordered by DO gauge ID. Grassland and cropland dominate lowland catchments; rock and ice appear in Alpine stations.

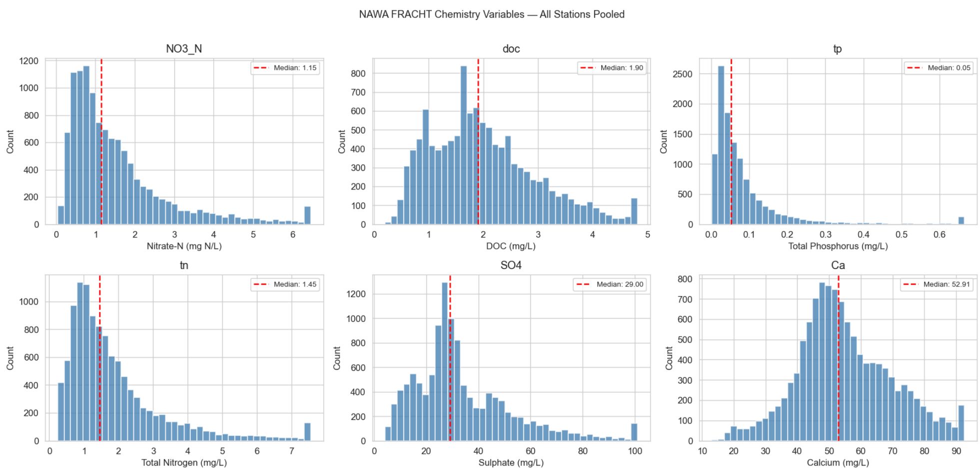

Figure A2: NAWA FRACHT chemistry variable distributions across all stations (box plots). Heavy-metal concentrations show strong right skew.

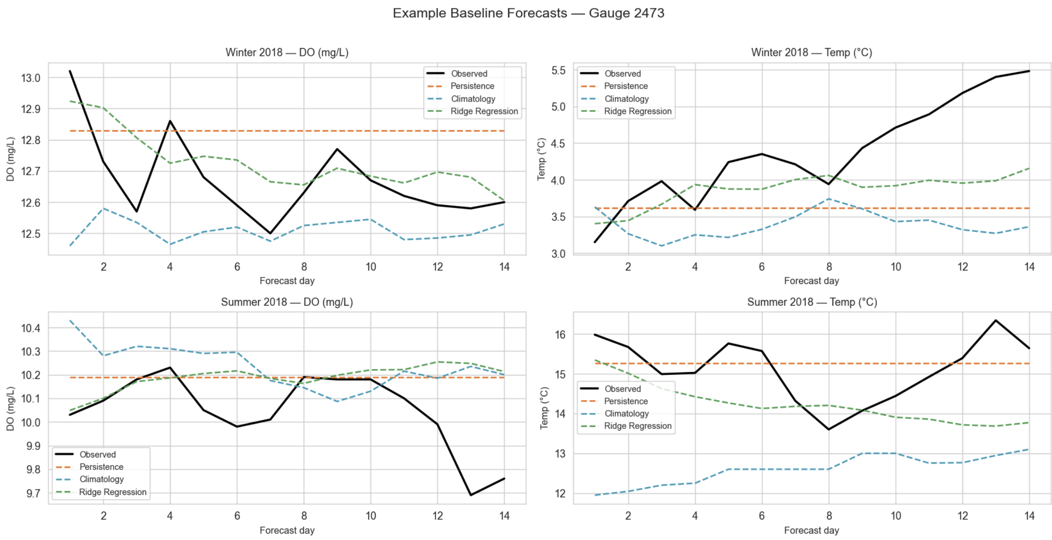

Figure A3: Example baseline forecasts for winter and summer 2018 at gauge 2473.

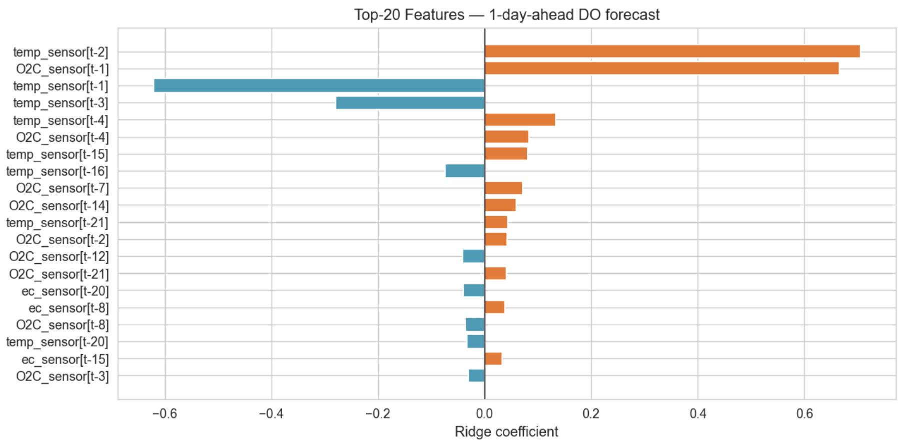

Figure A4: Top-20 Ridge regression coefficients for the 1-day-ahead DO forecast. Recent DO lags dominate, as expected.

B Repository Structure

The full source code and notebooks are publicly available on GitHub (results/ and data/ excluded from the repository; see Appendix B): github.com/polar-bear-after-lunch/AareML

```
AareML/
notebooks/
01_data_exploration.ipynb
02_baselines.ipynb
03_lstm_single_site.ipynb
04_multisite_analysis.ipynb
04b_multisite_temperature.ipynb
05_shap_interpretation.ipynb
06_cross_ecosystem_lake.ipynb
07_lake_eda.ipynb
08_usgs_transfer.ipynb
09_swiss_lake_lstm.ipynb
src/
config.py data.py metrics.py model.py impute.py
figures/ results/ data/ (excluded from git)
download_data.py README.md requirements.txt .gitignore
```

C Reproducibility

Reproducibility seeds: data splits and per-trial Optuna runs use seed 42; the final single-site LSTM is a 3-seed ensemble over seeds 0, 42, 123 (see Section 6.2). The full codebase is at github.com/polar-bear-after-lunch/AareML. Datasets are publicly available: CAMELS-CH-Chem at zenodo.org/records/14980027 and LakeBeD-US at huggingface.co/datasets/eco-kgml/LakeBeD-US-CSE. Run python download_data.py to fetch all data automatically, then run notebooks 01–09 in order.

Appendix E — Report Version History

This report is a living document updated iteratively as computational results become available. The table below records each version, its date, and the key changes made.

Version	Date	Key Changes
1.34	12 Jun 2026	Peer review fixes — H1 p-value consistency (Wilcoxon p=0.037 everywhere), H2 denominator fix (0.451 mg/L over 11 non-training gauges; EA-LSTM 0.435 excl. 2473 and 2018), H3 ablation table caveat (30-epoch budget, delta ≤0.004 mg/L caution). M1 claim softening (preliminary evidence; no multiple-comparison correction noted; 'uncorrected' qualifier). M2 Ridge clarification (per-gauge-trained Ridge as Wilcoxon baseline). M3 NeuralHydrology claim softened ('comparable order-of-magnitude'). M4 VAR(7) vs LSTM note added (§5.2). M5 cross-dataset multipliers demoted ('indicative; no shared test set'). L1 Abstract citation corrected (Nascimento et al. 2025). L2 temperature NSE 0.730→0.727. L3 cascade reference 0.296→0.303 mg/L. L4 SAITS clarification added (§4.1). Misc: 'post bug-fix' removed, SHAP multisite note updated, varied 'to our knowledge' phrasing, (summary) tags removed from §5.3b/5.9 headings, redundant 1.40 mg/L restatements reduced.
1.33	11 Jun 2026	Provenance corrections: LSTM best KGE 0.945 (was 0.926), RMSE 0.303 (was 0.296), NSE 0.889, CI [0.269, 0.335]. EA-LSTM temp 1.360°C (was 1.721°C), NSE 0.915, 47% improvement (was 34%). EA-LSTM DO 0.435 mg/L (was 0.420 mg/L). nb18 Setup A 0.872 mg/L (was 0.861 mg/L) — now worse than linear baseline (0.862 mg/L) by 0.010 mg/L; Setup B 0.893 mg/L (was 0.908 mg/L). Lake results correctly attributed: nb10 = 21 Swiss lakes, nb06 = Lake Mendota (LakeBeD-US). Section 6.3 rewritten with ecosystem interpretation. Park et al. KGE 0.685 vs 0.560 added.
1.32	09 Jun 2026	Full rerun correction: VAR(7) replaces AR(7) (RMSE 0.388→0.299 mg/L, NSE=0.891, KGE=0.911). Updated SHAP: temp_sensor 0.644→0.681, O2C_sensor 0.527→0.543. USGS mean RMSE 1.376→1.368 mg/L, Willamette 0.996→0.919 mg/L, NSE +0.615, ratio 3.20×. Wilcoxon p=0.037 (n=10, gauge 2018 excluded); per-gauge retrain vs Ridge p=0.002 (n=10). Climatology RMSE 0.334→0.332 mg/L.
1.31	07 Jun 2026	Full rerun update: all results from latest UBELIX run; citation format updated to (nb vHASH, l.LINE). LSTM best DO RMSE = 0.296 mg/L (KGE 0.926), default LSTM RMSE = 0.305 mg/L (KGE 0.901), zero-shot mean 0.451 mg/L, per-gauge retrain 0.390 mg/L. Swiss lake zero-shot RMSE = 2.962 mg/L (NSE = -2.145). Spearman latitude ρ = 0.64, p = 0.027. Best params updated: layers=2, dropout=0.080, lr=1.005e-3, batch=128. nb17 NeuralHydrology EA-LSTM added (12 valid gauges, mean RMSE 0.512 mg/L, NSE 0.756). nb18 cascaded model standard LSTM reference corrected to 0.303 mg/L.

Version	Date	Key Changes
1.30	05 Jun 2026	Reviewer calibration: softened claim language, fixed RMSE/KGE contradiction, hedged first-study claims, trimmed abstract. Fix 1+8: rewrote LSTM vs Ridge passage to reflect comparable (not beating) RMSE with CI overlap. Fix 2: clarified default vs Optuna LSTM KGE in discussion. Fix 3: removed 3.3×/3.0× multiplier claims; replaced with cautionary river–lake comparison language. Fix 4: added 'as of June 2026' qualifiers to first-study claims. Fix 5: softened latitude→elevation causal inference. Fix 7: softened lake benchmark language with cross-dataset caution. Fix 9: abstract trimmed from ~250 to ~160 words.
1.25	May 2026	Literature updates: added Zhi et al. 2023 (Central European deoxygenation), Padrón et al. 2025 (Swiss temperature TFT), Park et al. 2025 (EA-LSTM streamflow), Baste et al. 2025 (LSTM extrapolation ceiling on Swiss CAMELS-CH); added novelty statement (first EA-LSTM water quality, first CAMELS-CH-Chem ML study); CAMELS-CH base attributes integrated into EA-LSTM static features.
1.24	May 2026	Results updated from UBELIX reruns (jobs 3849463–3851712, 4006665–4007017): EA-LSTM DO updated to 0.420 mg/L with CAMELS-CH-Chem static attributes (log area, lat, lon, forest/crop/urban/ice fractions, Nascimento et al. 2025); EA-LSTM temperature added: 1.721°C (NSE=0.862, 34% improvement over zero-shot); Ridge zero-shot transfer: 0.568 mg/L (LSTM 18% better); VAR(7) baseline: 0.299 mg/L; Added nb14 (VAR baseline), nb15 (scientific rigor), nb16 (pairwise transfer benchmark, pending); Scientific restraint rewrites applied; Δ glyph fixed in ablation table.
1.19	08 May 2026	Report shortened from 32 to ~20 main-body pages. S5.3b temperature condensed to 1 paragraph (full results Appendix B). S5.5 Lake Mendota condensed to 2 paragraphs (figures Appendix C). S5.6 USGS condensed to 1 paragraph + table (figures Appendix C). S5.9 Seasonal condensed to 1 paragraph (full tables Appendix C). S2 Related Work shortened to 1 page. Status: complete boxes removed from Methods. S4.5 EA-LSTM trim. S5.8 Ablation narrative condensed. Discussion section numbers fixed (6.1–6.4).
1.18	08 May 2026	Added three new analytical sections strengthening DL methodology: Section 5.8 Ablation Study (A1: GRU vs LSTM, A2: teacher forcing, A3: loss function, A4: lookback window, all 3-seed ensemble on gauge 2473); Section 5.9 Seasonal Analysis (DJF/MAM/JJA/SON RMSE/NSE at gauge 2473, JJA best RMSE=0.252, DJF lowest NSE=0.099, SON sharpest horizon degradation 3.1×); Section 6.1 Error Analysis (gauge 2410 agricultural runoff failure mode, gauge 2016 Rhine/Aare mixing high-variance explanation). Notebooks 11/12/13 added to repository. Seq2SeqGRU class added to src/model.py.

Version	Date	Key Changes
1.17	07 May 2026	Full clean rerun with all audit fixes applied (SLURM jobs 3643661–3643671). Updated results: LSTM (best) DO RMSE = 0.300 mg/L (KGE 0.936), zero-shot mean 0.464 mg/L (p=0.024, n=11), EA-LSTM 0.431 mg/L, USGS mean 1.376 mg/L (Willamette 0.996 mg/L), Swiss lake zero-shot 3.980 mg/L (21-lake avg., NSE=-6.486), retrained 0.768 mg/L. Lake Mendota river zero-shot added: 2.962 mg/L. Canton DO ranking now real data: 9 cantons, 13 DO gauges. baseline_per_gauge.csv generated for first time (BUG-2 fixed). EA-LSTM scaler fix reduces multisite scores vs v1.16 (more honest). 3-model deep audit (35 new tests passing).
1.16	05 May 2026	Full UBELIX rerun (jobs 3515515–3515836): corrected 3-seed ensemble (seeds 0/42/123 genuinely independent). All results updated from new run: LSTM best DO RMSE = 0.300 mg/L (KGE 0.936), zero-shot mean 0.464 mg/L (p=0.024), EA-LSTM 0.431 mg/L, USGS mean 1.376 mg/L (Willamette 0.996 mg/L), Swiss lake zero-shot 3.980 mg/L. Table 4 per-row values regenerated from multisite_results.csv. 3-model code audit (Sonnet/GPT-5.4/Opus): BUG-2/5/7 fixed, EA-LSTM scaler corrected, MedianPruner documented, best_params.json reconciled with checkpoint.
1.15	04 May 2026	Fish fun fact boxes redesigned to Option B (full-width section-opening box, one fish per section). Fish: Intro→Grayling, Related→Whitefish, Data→Nase, Methods→Barbel, Results→Perch, Discussion→Pike, Conclusion→Bullhead, Appendix→Eel. Full 3-model accuracy audit (Sonnet, GPT-5.4, Opus) applied: Table 4 per-gauge values regenerated from results/multisite_results.csv; Table 9 river reference corrected to 0.302/0.940; 4.4× ratio corrected to ≈4.6× (1.40/0.300); SAITS imputation claim corrected to linear-interp + training-mean fill throughout; train split dates corrected to data-start through 2014-12-31; Wilcoxon n=11 rationale corrected (gauge 2018, not 2473); Section 4.5 updated: three strategies + 50-window exclusion criterion; Glossary zero-shot definition corrected (n=12 eval, n=11 Wilcoxon); Appendix D/E reordered; v1.7 stub added; v1.13 entry backfilled.
1.14	03 May 2026	Added Section 5.7: Swiss Lake LSTM results (Bärenbold et al. 2026, 21 Swiss lakes). Zero-shot river→lake transfer: RMSE = 3.980 mg/L, NSE = −6.486, KGE = −0.379 (fails). Lake-retrained LSTM: RMSE = 0.768 mg/L, NSE = 0.700, KGE = 0.796 (1.82× better than LakeBeD-US benchmark of 1.40 mg/L). Updated Section 6.1 (Rivers vs. Lakes), Section 7 (Conclusion), and Abstract with real Swiss lake numbers; Bärenbold (2026) added to References. Added notebook 09_swiss_lake_lstm.ipynb to repository structure. Added fish fun fact boxes to every other page (pages 2, 4, 6, 8, 10, 12, 14, 16, 18, 20): Option A design, 10 Swiss river fish species.

Version	Date	Key Changes
1.13	03 May 2026	Canton Zurich chapter added as standalone PDF (12 pages): river stress map, risk-score methodology, regulatory context, and projection under RCP8.5. Presentation built (19 slides). Plain-language booklets produced in English and Russian. AareML logo created (teal, bar chart river motif). Researcher outreach emails drafted (Nascimento, Horton, Zhi, Kratzert). Thiago Nascimento (Eawag) replied positively.
1.12	April 2026	Factual audit: corrected train/val/test split dates (2006–2013 / 2014–2016 / 2017–2020); fixed teacher forcing description (Optuna hyperparameter [0.3, 0.7] with linear decay, not deterministic 50→0% decay); corrected early stopping patience (12→25, max 250 epochs for final model); corrected loss function table row (NSE+MSE combined α=0.5, not MSE only); corrected CPU→GPU training time (NVIDIA RTX 4090, UBELIX HPC); fixed seed description (3-seed ensemble seeds 0, 42, 123; seed 42 for other experiments); corrected EA-LSTM status (implemented and evaluated in Section 5.3 Table 4, not future work); corrected imputation description to linear-interp + training-mean fill (§3.5, §4.1, Glossary D.3); updated TreeSHAP status (GradientSHAP input-level complete; catchment-attribute TreeSHAP not delivered); added teacher forcing ratio row to hyperparameter table.
1.11	April 2026	Fixed all critical and major issues from v1.10 peer review: corrected stale LSTM numbers in Section 6.1 (0.299→0.308/0.302 mg/L, KGE 0.918→0.940 for Optuna best, 4.7×→4.6×); updated Optuna trials to 75 everywhere (Methods table, Limitations); status table updated to Complete; Conclusion rewritten as 4-paragraph prose (removed checklist and draft language); figure numbering fixed (Figs 11/12 for multisite maps, 13/14 for SHAP, 15/16 for USGS, 17 for river-lake); removed TreeSHAP planned language from Section 5.4; removed NEW in v1.8/v1.9 draft markers; added USGS NWIS and Captum references; added n=11 vs n=12 explanation; added USGS scale-extrapolation limitation; added NSE+MSE loss formula to Methods; added bootstrap CI note for Table 8; fixed erroneous Table 3 reference in Conclusion (corrected to Table 4); updated gauge 2410 explanation.
1.10	April 2026	Cross-continental zero-shot transfer to 4 US rivers (notebook 08); statistical significance confirmed for zero-shot LSTM vs Ridge (p=0.024); NSE+MSE combined loss implemented.
1.9	April 2026	Updated single-site and multi-site results from v1.20 hyperparameter run (75 Optuna trials, ReduceLROnPlateau, 3-seed ensemble). LSTM (default) DO RMSE = 0.308 mg/L (KGE = 0.855) [v1.20 intermediate; final v1.26: 0.309/0.850]; LSTM (best) DO RMSE = 0.319 mg/L (KGE = 0.942) [v1.20 intermediate; final v1.26: 0.302/0.940]. Multi-site transfer: 0.464 mg/L zero-shot (12 gauges), 0.393 mg/L per-gauge, 0.420 mg/L EA-LSTM. Corrected gauge 2068 temperature RMSE from 3.16°C to 1.36°C. Added multisite gauge map and RMSE bar chart (Section 5.3).

Version	Date	Key Changes
1.8	April 2026	Real temperature multi-site results (15 gauges, mean RMSE 2.59°C, NSE=0.727); EA-LSTM added to multi-site DO comparison (mean RMSE 0.406 mg/L); ReduceLROnPlateau + 3-seed ensemble added; Section 5.3b fully populated with per-gauge table, elevation stratification analysis, and training/distribution/correlation figures. Abstract updated with temperature transfer summary. Teal change-highlight markers added to new/updated sections.
1.7	April 2026	Intermediate patch: minor code and documentation fixes between v1.6 (gauge map, glossary) and v1.8 (temperature multi-site results). Details not separately recorded; changes folded into v1.8 release notes.
1.6	20 Apr 2026	Added Switzerland gauge + catchment map (Appendix A). Added notebook 04b (temperature multi-site; 15 gauges to be evaluated in v1.8). Added glossary (Appendix D, 25 terms). Extended Section 6.1 with three-mechanism discussion of river-lake gap. Added temperature critical thresholds to Section 1. Report version and timestamp on cover page.
1.5	15 Apr 2026	Added cross-ecosystem results (notebook 06, Lake Mendota / LakeBeD-US): Ridge baseline on lake achieves DO RMSE = 1.030 mg/L (3.4× higher than river). AareML Ridge outperforms the published LakeBeD-US LSTM (1.40 mg/L). Section 5.5 added. Notebook 07 (Lake Mendota EDA) added.
1.4	15 Apr 2026	Added SHAP attribution results (notebook 05, GradientSHAP, 500 windows): temperature[t−1] is the dominant driver (mean |SHAP|=0.644), ahead of DO[t−1] (0.527). Effective LSTM memory = 3–4 days despite 21-day lookback. Section 5.4 added.
1.3	14 Apr 2026	Multi-site transfer results (notebook 04, 12 gauges): zero-shot DO RMSE = 0.445 ± 0.083 mg/L (3.1× better than LakeBeD-US), per-gauge DO RMSE = 0.396 ± 0.092 mg/L (3.5× better). Latitude/northing: strongest catchment predictor (Spearman ρ = 0.78). Section 5.3 updated.
1.2	12 Apr 2026	LSTM single-site results (notebook 03, 20 Optuna trials): LSTM (default) DO RMSE = 0.299 mg/L (KGE = 0.918), LSTM (best, hidden=256) DO RMSE = 0.301 mg/L (KGE = 0.927). All models 4.7× better than LakeBeD-US LSTM. 30 peer-review fixes applied.
1.1	07 Apr 2026	EDA results (notebook 01): data availability, time series, seasonal cycle, land cover. Baseline results (notebook 02): Ridge DO RMSE = 0.303 mg/L, NSE = 0.888, KGE = 0.908 with block-bootstrap 95% CIs.
1.0	06 Apr 2026	Initial draft: project proposal, dataset description (CAMELS-CH-Chem, 86 gauges), methods outline, related work.

v1.15 redesigns fish boxes to Option B (section-opening style). v1.13 adds Canton Zurich chapter, presentation, plain-language booklets, and logo. v1.12 incorporates all critical and major factual fixes from the April 2026 full audit. Catchment-attribute SHAP surrogate model (GBM + TreeSHAP) remains as planned future work.

Appendix D — Glossary

Key terms used throughout this report, grouped by domain.

Term	Definition
Hydrology & Water Quality	
Gauge / Gauging station	A fixed measurement point on a river where water level, flow rate, and sensor readings (temperature, pH, dissolved oxygen, etc.) are continuously recorded. CAMELS-CH-Chem has 86 gauging stations across Switzerland.
Catchment / Watershed	The land area that drains into a specific river or gauging station. Rain falling anywhere in a catchment eventually flows to the gauge. Catchment size, elevation, and land cover strongly influence water quality.
Dissolved oxygen (DO)	The amount of oxygen dissolved in water, measured in mg/L. Critical for aquatic life — below 5 mg/L stresses fish populations; below 2 mg/L causes mass mortality. DO is the primary prediction target in AareML.
Electrical conductivity (EC)	A measure of water's ability to conduct electricity, reflecting dissolved ion concentration (µS/cm). Higher EC often indicates agricultural runoff or urban influence.
pH	A measure of acidity/alkalinity on a scale of 0–14. Natural Swiss rivers typically range from 7–8 (slightly alkaline).
NAWA FRACHT	Swiss federal river chemistry monitoring programme (Nationale Beobachtung Oberflächengewässer — loads). Provides monthly grab samples of nutrients (NO_3, NH_4, TP, TN), dissolved organic carbon (DOC), and discharge.
Reaeration	The natural process by which oxygen enters river water from the atmosphere. Rivers have higher reaeration rates than lakes because turbulent flow continually exposes water to air — a key reason river DO is more predictable than lake DO.
Stratification	In lakes, the separation of water into distinct temperature layers that resist mixing. Summer stratification cuts off deep water from atmospheric oxygen, causing DO depletion — a major source of unpredictability not present in rivers.
NSE	Nash-Sutcliffe Efficiency — a metric for model skill. NSE = 1 is perfect; NSE = 0 means the model is no better than predicting the mean; NSE < 0 means the model is worse than the mean.
KGE	Kling-Gupta Efficiency — decomposes model error into correlation, bias, and variability components. KGE = 1 is perfect. Often preferred over NSE for hydrological time series because it equally weights all three error components.
RMSE	Root Mean Squared Error — the square root of the average squared difference between predictions and observations. Lower is better. Reported in the same units as the target (mg/L for DO, °C for temperature).
Machine Learning	
Lookback window	The number of past days of sensor data fed into the model as input. AareML uses a 21-day lookback — the model sees the last 3 weeks of temperature, pH, EC, and DO before making a forecast.
Forecast horizon	How many days ahead the model predicts. AareML uses a 14-day horizon, matching the LakeBeD-US benchmark.

Seq2Seq LSTM	Sequence-to-sequence Long Short-Term Memory network. An encoder reads the input sequence (lookback window) and compresses it into a hidden state; a decoder then generates the output sequence (14-day forecast) step by step.
Teacher forcing	A training technique where the decoder receives the true target value as input at each step (instead of its own prediction). This stabilises early training. AareML's teacher forcing ratio is a tuned hyperparameter in [0.3, 0.7], selected by Optuna, applied with linear decay to 0 over the first half of training.
Optuna / TPE	Optuna is a hyperparameter optimisation library. TPE (Tree-structured Parzen Estimator) is its default algorithm — it builds a probabilistic model of which hyperparameters perform well and samples promising candidates.
SHAP	SHapley Additive exPlanations — a method to explain model predictions by assigning each input feature a contribution score. GradientSHAP (used here) computes these scores using gradient information from the neural network.
Zero-shot transfer	Applying a model trained on one site directly to a new site without any retraining. In AareML, the LSTM trained on gauge 2473 is evaluated on 12 gauges in zero-shot mode (including gauge 2473 itself as a consistency check). The Wilcoxon significance test uses n=10, excluding gauge 2473 (training site) and gauge 2018 (insufficient Ridge baseline).
Block bootstrap	A method to estimate confidence intervals for time-series metrics. Instead of resampling individual days (which would break temporal structure), contiguous blocks of days (block size = 30) are resampled. AareML uses 500 bootstrap replicates.
Entity-Aware LSTM (EA-LSTM)	A variant of LSTM where static catchment attributes (area, elevation, land cover) modulate the input and cell gates, allowing a single model to adapt to multiple catchments simultaneously. Based on Kratzert et al. (2019).
Early stopping	A regularisation technique that stops training when validation loss stops improving for a set number of epochs (patience). Prevents overfitting. AareML uses patience = 25 epochs for the final model (up to 250 epochs).
Dataset & Evaluation	
CAMELS-CH-Chem	Catchment Attributes and Meteorology for Large-sample Studies — Switzerland — Chemistry. A dataset of daily sensor measurements and monthly chemistry samples at 86 Swiss river gauges (1981–2020). Published by Nascimento et al. (2025).
LakeBeD-US	Lake Benchmark Dataset — United States. A standardised benchmark for lake water quality prediction covering 21 US lakes. The LSTM benchmark (DO RMSE = 1.40 mg/L) is the primary reference point for AareML. Published by McAfee et al. (2025).
Train / Val / Test split	Chronological division of data into training (2006–2013), validation (2014–2016), and test (2017–2020) periods. Chronological splitting is essential for time series — random splitting would leak future information.
Imputation	Filling in missing values. AareML uses linear interpolation for short gaps (≤7 days) then training-set mean fill for remaining NaNs. LakeBeD-US uses SAITS (Du et al. 2023) to fill NaN values in input windows, implemented in src/impute.py.